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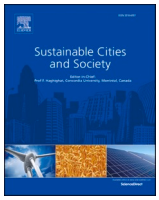

A method of estimating the spatiotemporal distribution of reflected sunlight from glass curtain walls in high-rise business districts using street-view panoramas

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ABSTRACT

Obtaining the time and location of sunlight reflection within a street environment is important to improve the comfort and safety of public spaces. This paper proposes a method to estimate the spatiotemporal distribution of sunlight reflection within a street based on street-view data. First, the solar path on the panorama is plotted to obtain the potential orientation of the radiation source and the reflection on building facades. Next, the continuous time–date distribution of direct and reflected sunlight at a certain site is obtained by the panoramic projection transformation approach. The accumulative duration and time-series parameters of the reflected scenario are estimated by a matrix operation. By mapping the time-specific duration of the sunlight reflection scene, we can obtain the spatiotemporal distribution within the street network. We applied the proposed method to a case study in a high-rise business district and quantified the range, duration, and intensity of reflected sunlight in the surrounding area. Seasonal changes, city morphology, and facade orientation were found to be the factors affecting the distribution tendency of reflected sunlight. The risk detection, visualization, and quantification analysis results on sunlight reflection can provide a reference for city management, urban planning, and environmental evaluation.

1. Introduction

1.1. Background

Urban heat has become a worldwide climate crisis (Levermore, Parkinson, Lee, Laycock & Lindley, 2018; Sanchez-Guevara, Núñez Peiró, Taylor, Mavrogianni & Neila González, 2019). The impacts of high ambient temperature on public health (He et al., 2022; Mastrangelo, Visentin, Milan, Fadda & Spolaore, 2007), air quality (Lai & Cheng, 2010; Ulpiani, 2021), and thermal comfort (He, Wang, Liu & Ulpiani, 2021b; Shashua-Bar, Pearlmutter & Erell, 2011) have been widely recognized. Many efforts have been made to reduce the impact of urban heat in the reduction of solar radiation absorption (Aleksandrowicz, Vuckovic, Kiesel & Mahdavi, 2017; He et al., 2021a) and the enhancement of latent heat release (Mirzaei & Haghighat, 2010; Zhao, He, Li, Wang & Darko, 2017). Among the policies to reduce urban heat, increasing the urban albedo is considered a potential means of alleviating the absorption of solar radiation. (Akbari, Damon Matthews & Seto, 2012; Wang, Berardi & Akbari, 2016). The mitigation potential of reflective materials depends on a set of factors, including material characteristics (Rosso, Pisello, Cotana & Ferrero, 2016), building characteristics (Falasca et al., 2019), and urban morphology (Yang, Wang & Kaloush, 2015). The application of reflective materials in pavements or wall surfaces is thought to be less desirable (Erell, Pearlmutter, Boneh & Kutiel, 2014). Although the use of reflective materials, such as glass curtain walls (GCW), helps to increase the urban albedo (Taleghani, Swan, Johansson & Ji, 2021), it also increases the mean radiant temperature at the street level (Salvati et al., 2020). Moreover, sunlight reflection caused by reflective facades leads to the uneven distribution of radiation absorption within street canyons, which has an adverse impact on latent heat release (Schwarz, Schlink, Franck & Großmann, 2012; Vasilakopoulou, Kolokotsa & Santamouris, 2014). In addition to considering thermal comfort, visual comfort should be taken into account when applying cool materials at the city scale (Falasca et al., 2019). Reflective materials may produce visual discomfort in sunny weather conditions, which significantly affects pedestrians' visual

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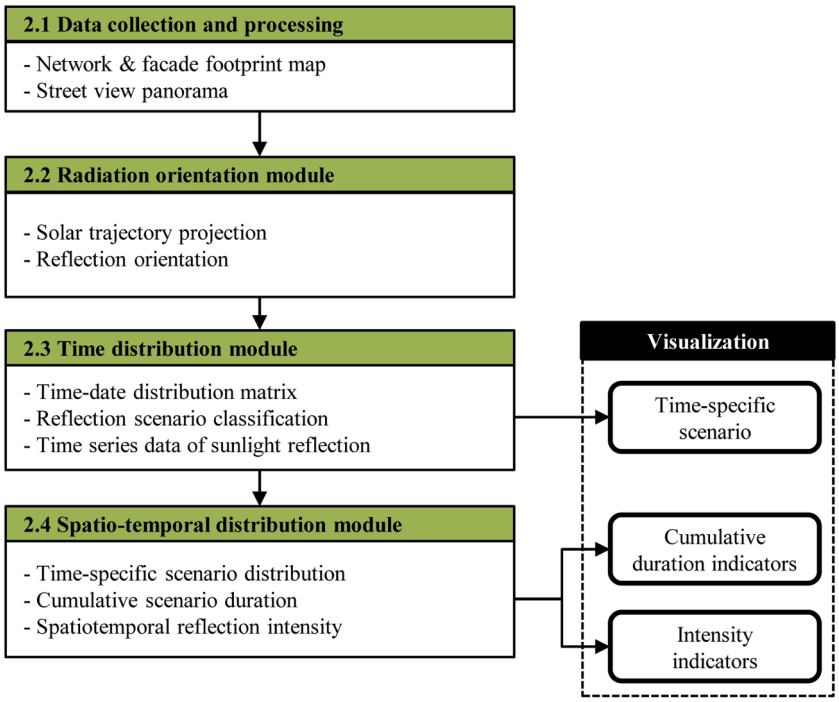

Fig. 1. Model Framework.

perceptions (Rosso et al., 2016). Hence, controlling the reflected solar radiation is important for the improvement of street-level microclimate.

Reflected sunlight is a common phenomenon in cities. It is caused by smooth surfaces that exist widely in our daily life, such as plane pavement, windows, building facades, windshields, stationary water, photovoltaic panels Hayward (2012), and even art installations Markusoff (2014). Sunlight from a GCW façade is a type of reflection that has negative impacts on the safety and comfort of the urban environment (Danks, Good & Sinclair, 2016; Gelan, 2012; Zhao, Arshad, Wang & Triantafilis, 2021). Damage may also be done to people and cars due to focused sunlight from glass coverage and curvilinear design facades Schiler (2009), while accidents may be indirectly caused by sudden apparent glare during driving (Grey & Regan, 2007). Moreover, the increased exposure to sunlight affects people's cognition of thermal comfort (Danks et al., 2016).

1.2. *Dynamic impact of reflection*

Sunlight reflection is not a stationary phenomenon occurring at a specific location. Reflection from a GCW façade of high-rise buildings generally has a wider-impact region in street canyons and open spaces (Wong, 2015; Yang, Grobe & Stephen, 2013). Reflected irradiation continuously reaches different sites in the surroundings, along with solar movement in the sky (Shih & Huang, 2001). A long reflection duration has a significant effect on the microclimate in street canyons (Jin, Kang, Jin & Zhao, 2017; Schiler, 2005). Quantification of the impact duration of reflected sunlight is important for analyzing people's exposure time and thermal discomfort (Danks et al., 2016), as it provides indicators for evaluating thermal impacts on specific time periods (LAI, 2012; Parsons & Ken, 2006). The effect of reflected sunlight on the street environment is time-specific (Aghamolaei, Fallahpour & Mirzaei, 2021). Solar radiation reflected by a GCW is considered as thermal discomfort in the hot summer Gelan (2012) and thermal comfort to shadow regions in cold winters. Therefore, the spatiotemporal distribution of reflected sunlight is an essential issue for comprehensively evaluating the GCW façade reflection's thermal impact to the street environment. In this paper, a new method is proposed to quantify the time-series indicators of the duration and intensity of daylight reflection in street space.

Reflected sunlight has a negative visual impact on urban environments. Compared to sunlight reflection on other types of surfaces, reflection on a GCW is more serious owing to the vision disability and discomfort caused by a dazzling glare in a view field (Schieber & Harms, 1998). For drivers, reflected glare is generally unexpected compared to direct solar glare. Sudden glare apparent in a view field may degrade driver performance (A. M. City, 2015; Grey & Regan, 2007). In a commercial region with a high density of high-rises, wide usage of GCWs increases the apparent possibility of glare in the street environment. The sites' large amount of existing reflective surfaces also carries a high risk of several sunlight reflections occurring at the same time (Rofail, Dowdle & Perry, 2004). Bright light from different directions will occupy a certain area of the view field, leading to the temporary vision loss of pedestrian and drivers (Grey & Regan, 2007). The conditions of reflection focus are more complicated and vary with the facades' orientation and solar position Gil (2015). It is generally difficult to detect the time and location of reflection focus scenarios in a commercial region. Obtaining the time-specific condition of sunlight reflection from GCW facades is important for preventing and reducing glare damage. The method proposed in this paper can be used to identify risky scenes of multiple dazzling light sources in urban street environment and to quantify the cumulative impact duration of the scenes overall the year.

1.3. *Models and tools*

Many methods, models, and software have been developed to measure, estimate, and simulate solar reflection, both in the indoor (Wienold & Christoffersen, 2006) and the outdoor environment (Shih & Huang, 2001).

Generally, the 3D spatial model of buildings, terrain, and the solar path in the urban environment must be preset to analyze sunlight reflection in a limited region (Yang et al., 2013). There are limited models and research on the comprehensive effect of reflected irradiation from multiple curtain walls in a high-rise region. For analyzing reflection's spatial distribution in the city or regional range, a large amount of urban spatial data are necessary Wong (2015). In addition, estimating the reflection impact area on a large time and spatial scale requires huge calculation. Furthermore, actual optical and thermal

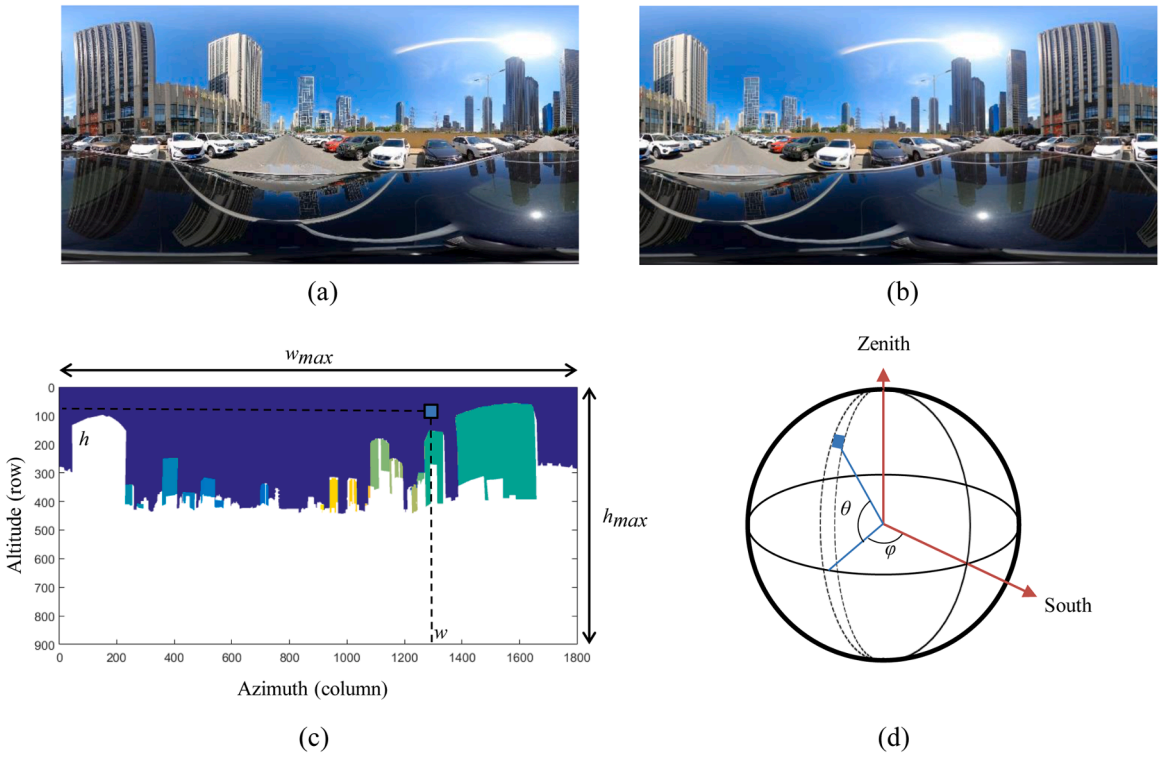

Fig. 2. Panorama image processing: (a) initial image, (b) rotated image, (c) sky and façade extraction result, (d) spatial orientation coordinates.

conditions within a micro urban space were affected by dynamic and detail environmental factors (e.g., trees, shading facilities, and billboards), which limited the accuracy of simulation results. It is generally difficult to apply a simulation method to quantitatively examine the performance of sunlight reflection prevention measures. This study provides a novel method to detect the risk scene in a thermal and optical environment on an urban scale based on limited digital data and actual environmental information.

Fortunately, street view images provide a possible means to obtain the spatial relation of perceived environmental information through panoramic projection (Kang, Wang, Körner, Taubenböck & Zhu, 2018; Kim, Lee, Hipp & Ki, 2021). The spatial orientation and solid angle of environmental objects can be obtained by the position and area of image pixels Ohno (2018). Google Street View panoramas have been widely applied to conduct quantitative (Richards & Edwards, 2017; Yang, Ao, Ke, Lu & Liang, 2021) and automatic (Yang et al., 2020; Zhai & Peng, 2020) analysis on street environmental characteristics in the city range. Hemispherical images provide a good supplement for the simulation of solar radiation and environmental evaluation within street canyons (Chen et al., 2012; Liu et al., 2019). Li and Ratti, 2019 proposed a method of estimating the spatiotemporal solar radiation within street canyons based on Google Street View panoramas and the building-height model. However, more attention is paid to the quantitative measurement of direct and diffuse irradiation; there are few models considering reflected irradiation. Ying et al. (2021) applied panorama segmentation to estimate the spatial distribution of reflected sunlight on extracted GCW facades on several sample times on the spring equinox. Quantification of the reflection duration was not involved. The method proposed in this study quantifies the hour-by-hour variation of reflection intensity.

1.4. Research aim

The objective of this paper is to present a method to detect and quantify the dynamic sunlight reflection in the urban space. This method does not rely on the 3D model but the street-view panorama, which is easy to purchase. This method adopts spatial sampling to clarify the time distribution of reflection scenes at different locations so as to obtain the time-series influence range of reflective facades in the urban space. Through reflection scenario classification and spatial parameter calculation, we aimed to purchase data describing the reflection condition in detail. These data were expected to provide indictors for evaluating reflective facades' impact on thermal and visual comfort. Based on the quantitative measurement of reflection duration and intensity, we visualized the spatiotemporal distribution of reflected sunlight in order to provide a reference for urban planning, landscape design, and smart city management.

2. Methodology

Fig. 1 shows a scheme of the proposed method. Section 2.1 introduces how to collect and process required data. Section 2.2 describes the approach of estimating the reflected sunlight's orientations and their corresponding sun angles at a specific site. Section 2.3 describes how sun angles on the panorama are transformed into the time–date coordinates, and how the time distribution of direct and reflected radiation is taken as output. Section 2.4 defines site-specific indicators to visualize the spatiotemporal distribution. The spatiotemporal distribution module uses the time-series result of reflection scenarios at different sample sites to illustrate the time-specific and accumulative distribution tendency of sunlight reflection in the study area.

2.1. Data collection and processing

2.1.1. Data collection

The aim of the proposed method is to obtain the dynamic of sunlight reflection at the street-level urban space. The model input requires the surface orientation data of GCW facades and the street-view panoramas taken at a set of sample sites homogeneously located on urban street networks within the study area.

The facade orientation data can be obtained by the footprint map of GCW facade buildings from open-source data (e. g. OpenStreetMap). We

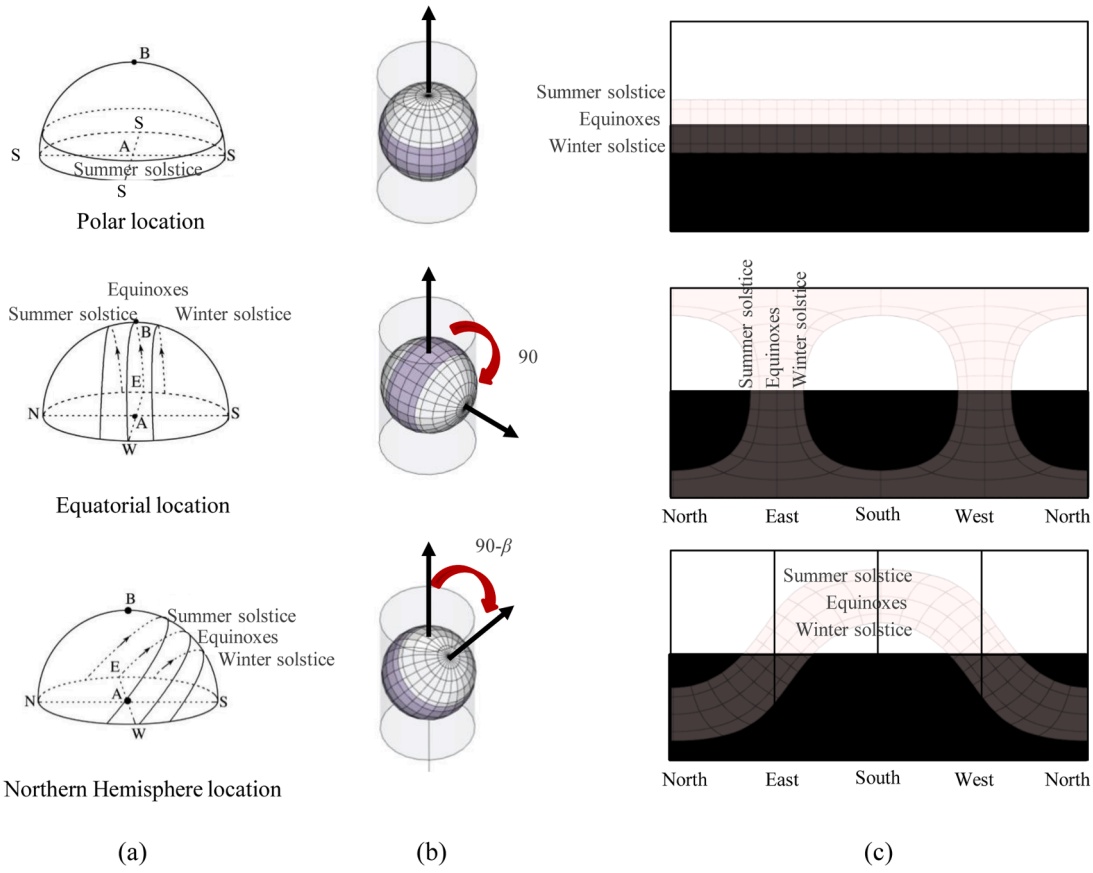

Fig. 3. Panorama projection of solar trajectories at different altitudes: (a) solar path, (b) panoramic projection, (c) panorama.

use polygons to represent the high-rise buildings' facade data. Each polygon edge standing for a GCW facade is isolated to a specific layer with a unique index.

The proposed model represents the street-level urban space by a link–node network. Nodes are used to represent the center of road intersections, while links standing for urban streets are generated based on the middle lines of road. Sample sites are generated with a certain interval along the links in the network. For other shapes of urban space, sample sites can be generated base on the mesh grid with certain intervals. The sampling intervals are decided by the scale of the study area and the spatial granularity of analysis.

The street-view panorama data can be obtained through field investigation or open-source data. As the data collection through street view API has been introduced in many previous studies, this paper gives a field investigation method to collect street-view panoramas in areas lacking data. The continuous street-view data along the links are recorded by a panoramic video camera fixed on top of a car moving along the sample streets. The frames abstracted from the panoramic video are dealt with by a calibration approach to obtain the position coordinates and front orientation of street-view data. The calibration process is introduced in Appendix B. The panorama images at the sample sites are exported as the initial data for analysis.

2.1.2. Image processing

For each sample site, the street-view panorama is exported in the format of an equidistant cylindrical projection (Fig. 2(a)). We take the horizontal axis as the azimuth and the vertical direction as the zenith. All panoramas are rotated until the south in centered in the image (Fig. 2 (b)).

Sky and GCW extraction is a prerequisite step for estimation. To ensure the accuracy of the result, the manual classification method is applied to obtain the potential source of reflected sunlight in the street view. As shown in Fig. 2(c), sky pixels are first extracted and identified in a specific color. Each GCW surface and its potential reflection region are extracted manually and filled with a color corresponding to the ID of the facade. The extraction method of the potential reflection region on a visible GCW facade is given in Appendix C. The segmentation result of each image pixel can be obtained by its RGB tunnel values, which index a specific facade ID. As shown in Fig. 2(c), the segmentation results are exported by images with a resolution of 1800 × 900 pixels in this study. For each pixel in the processed panorama, its spatial orientation (φ, θ) in the viewpoint of the sample site (Fig. 2(d)) can be obtained by its image position (w, h).

$$\varphi = \frac{2\pi}{w_{max}}(w - w_{max}/2), \tag{1}$$

$$\theta = \frac{\pi}{h_{max}}(-h + h_{max}/2), \tag{2}$$

where w_{max} and h_{max} stand for the width and height of the panorama image, respectively.

2.2. Radiation orientation module

2.2.1. Solar trajectory projection

This module adopts a projection conversion approach to obtain the orientation of beam radiation based on solar trajectories on the panorama. As shown in Fig. 3(a), solar motion trajectories in the sky, which are represented by a hemisphere, occupy a belt-shaped area varying with location altitude. If we ignore the occlusion of buildings and the earth and represent it with the celestial coordinate system, the 1-year solar-apparent motion trajectory at the altitude β is composed of 1-day circle trajectories with a rotation axis of (90-β) obliquity (Fig. 3(b)). The difference of trajectories at solstice and equinox is equal to the

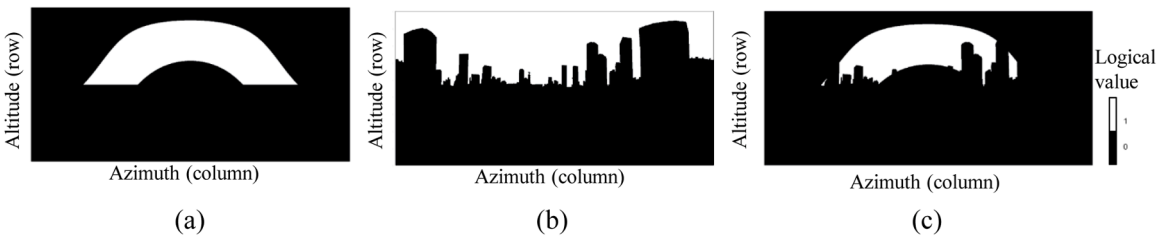

Fig. 4. Spatial orientation estimation of direct solar radiation in street-view panorama: (a) solar trajectory matrix, (b) sky segmentation matrix, (c) direct sunlight orientation matrix.

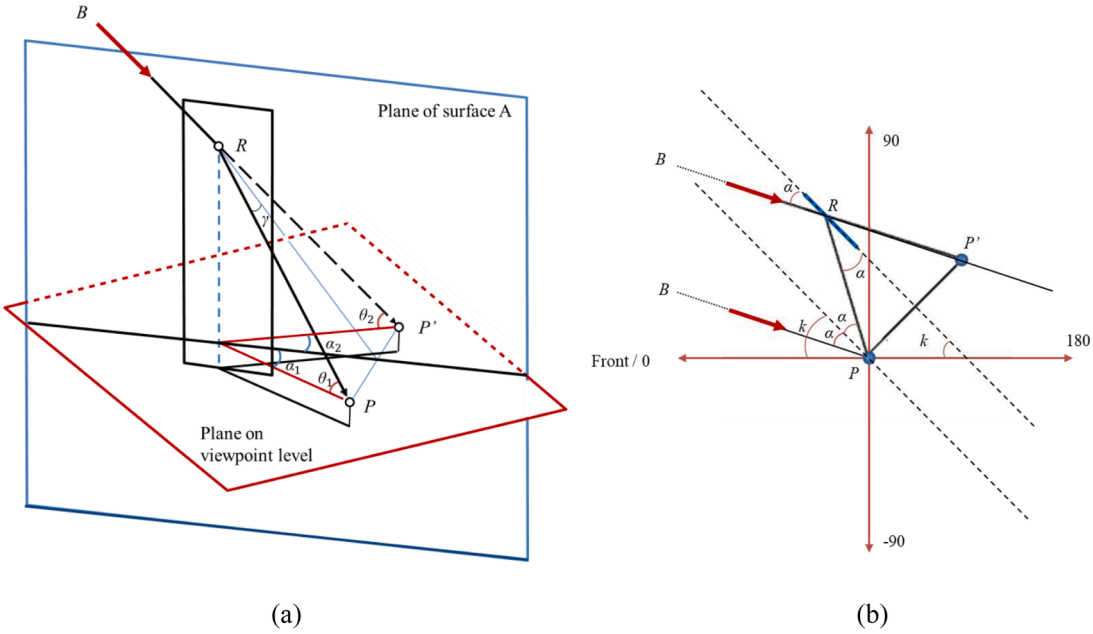

Fig. 5. Sunlight reflection on plane surface: (a) 3-dimensional view, and (b) 2-dimensional view.

earth's axial tilt.

Using cylindrical projection, Fig. 3(c) plots the panoramic solar trajectories at different altitudes on the panorama. For the panoramic projection at polar points, the normal directions of 1-day trajectories are parallel to the central axis of the projection cylinder. The trajectories of other latitudes can be considered as the panoramic projection of the celestial sphere rotted with β degrees around the east–west axis. Hence, panoramic solar trajectories at the latitude β of the geographical location can be obtained by panoramic projection conversion. The details on the conversion approach are described in Appendix D.

The result of solar trajectory projection for all simple sites in the study area are denoted by a 900×1800 logical matrix, *SR*, with the element of 0 and 1. In Fig. 4(a), the white area represents the possible orientation of direct solar radiation throughout the year by the element of 1.

The sky extraction result of the sample site *s* is also denoted by a 900 × 1800 logical matrix, *KR*. The spatial orientation of direct sunlight, *DR*...., is estimated by the intersection of the visible sky region and the solar path (Fig. 4), as follows:

$$\mathrm{DR}....= \mathrm{KR}. * \mathrm{SR}. \tag{3}$$

The cells of 1 in the matrix *DR*.... (white area in Fig. 4(c)) indicate the spatial orientation of direct sunlight accessible at the sample site throughout the year.

2.2.2. Sunlight reflection orientation

The generation of reflected sunlight from the GCW is affected by factors such as the direction of sunlight incidence, the location of the interface space, and the surface orientation. Assume that a location of altitude β exists on a vertical reflective surface, *A* (Fig. 5(a)). One beam of sunlight from the sun, *B*, reaches the viewpoint, *P*, through the reflection of *A*. For point *P*, the glare image appears at point *R* on the background of A. The incident and reflection direction can be represented by the arrays $\overrightarrow{\mathrm{BR}}$ and $\overrightarrow{\mathrm{RP}}$.

Based on the plane of surface *A*, we can obtain the mirror point, *P'*, of the viewpoint. According to the rule of mirror reflection, we can obtain the equivalence of angles as follows:

$$\theta_1 = \theta_2, \tag{4}$$

$$\alpha_1 = \alpha_2, \tag{5}$$

where θ_1 and θ_2 are the angles between the incident/reflection light and horizontal plane, and α_1 and α_2 are the angles with the plane of surface *A* (Fig. 5(b)). As *P'* is located on the extension of array $\overrightarrow{\mathrm{BR}}$, the refection point, *R*, can be obtained by the intersection of segment *BP'* and surface *A*.

Taking the south direction as the front, we can construct a spherical coordinate system to plot the scene in the viewpoint of *P* on the panorama (Fig. 6(a)). As *B* is far enough from location *P*, we can approximately consider array $\overrightarrow{\mathrm{BR}}$ as parallel to array $\overrightarrow{\mathrm{BP}}$. Using a two-dimensional Cartesian coordinate system (Fig. 6(b)), we can find that the azimuths of direct and reflection sunlight should be

$$\varphi_1 = k - \alpha, \tag{6}$$

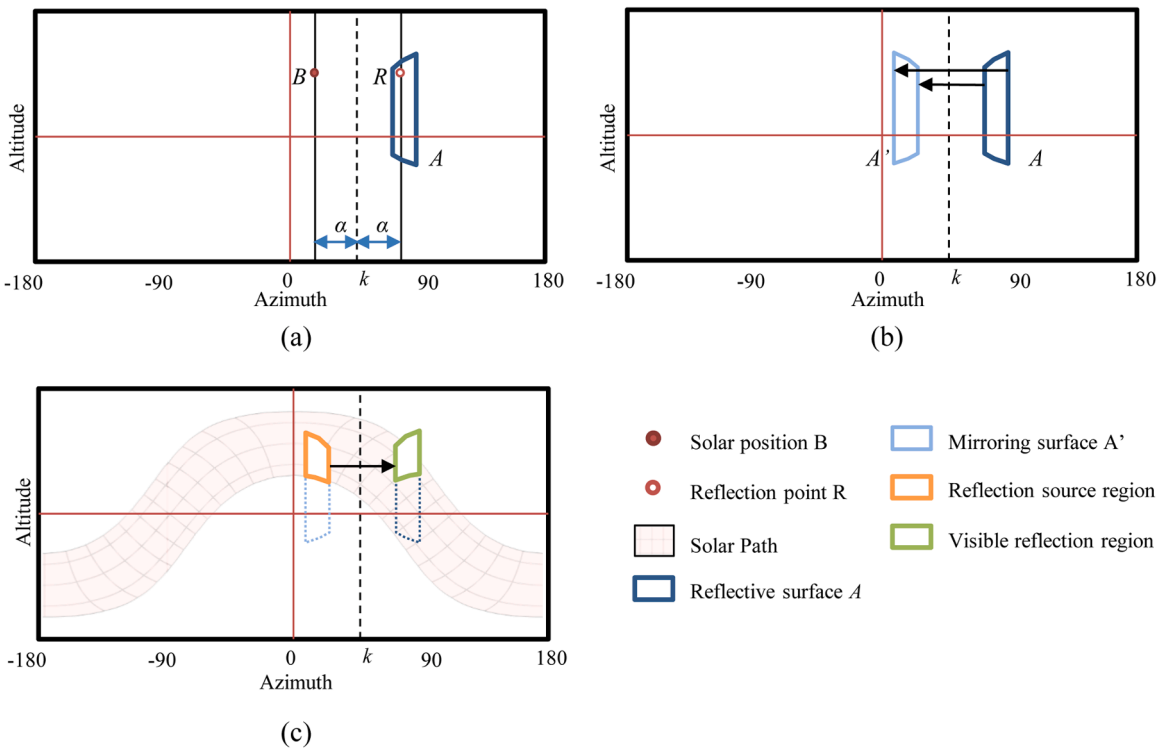

Fig. 6. Spatial orientation estimation of reflected sunlight and reflection source region on panorama.

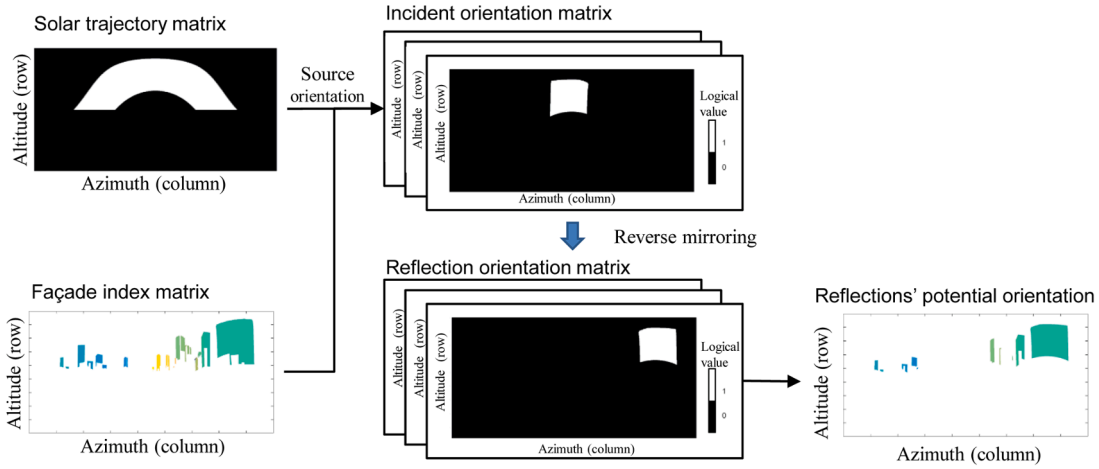

Fig. 7. Estimation approach of reflection source region and visible reflection region.

$$\varphi_2 = k + \alpha, \tag{7}$$

where k is the angle between the south and surface A's projection on the plane. According to function (6) and (7), the positions of sun B and the reflection point R should be symmetrical in the panorama based on the surface direction, k (Fig. 6(a)).

For each reflection point on surface A, its incidence direction of sunlight can be obtained by its mirroring point along the surface direction k. Namely, we can obtain the potential orientation of the radiation causing the reflection on surface A based on its mirroring area, A' (Fig. 6(b)). This region is called the reflection source region in this study.

The overlapped area of A' with the solar path indicates the solar direction that may cause reflection sunlight on the corresponding position on surface A (Fig. 6(c)). However, the potential region of reflection point, R, on surface A can be obtained by reversely mirroring the reflection source region. The reverse mirrored area is called the visible reflection region in this study.

For each facade $i \in I_s$ visible at the sample site, its segmentation result is denoted by an image-size 0–1 matrix, GR_u. The incident orientation of the radiation can be obtained by mirroring the cells in the matrix GR_u based on the facade orientation column, k_u.

$$MR_u(w,\ h) = GR_u(2k_u - w,\ h),\ \forall w,\ \forall h, \tag{8}$$

where w and h stand for the cell's position in the matrix. The intersection between the potential source region and the solar trajectories is obtained by matrix operation as follows:

$$OR_u = MR_u . * SR, \tag{9}$$

where OR_u is a facade-specific logical matrix indicating the solar orientation causing reflection on the facade u. The spatial orientation of

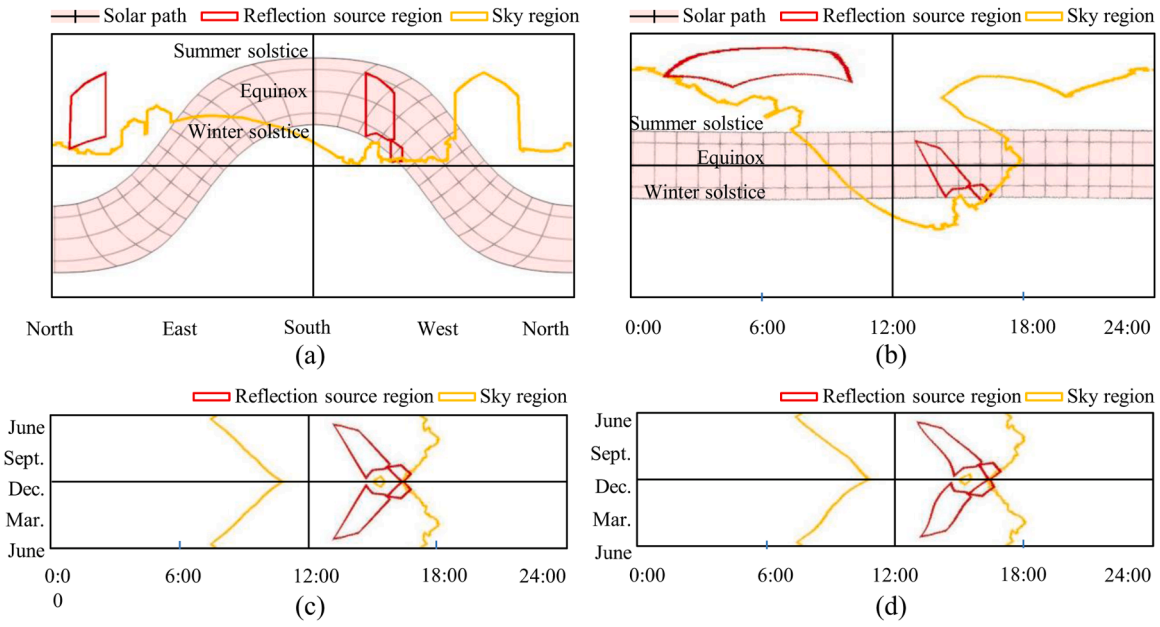

Fig. 8. Estimating time distribution through projection transformation: (a) time scale on the street-view panorama, (b) time scale after transformation, (c) date coordinate calibration, (d) and EQT calibration result.

sunlight reflection can be obtained by the mirrors of OR_u based on the facade orientation. Repeating the above approach for all visible facades at the sample site, we can obtain the possible orientation of sunlight reflection on the panorama. (Fig. 7)

This module outputs the solar angle and the facade-specific incident orientation data by the 0–1 matrixes of the same size with the panorama image.

2.3. Time distribution module

2.3.1. Direct and reflected radiation duration

As mentioned above, the sun trajectories on the panorama of altitude β are obtained by projection transformation. Accordingly, for each pixel within the solar path, its corresponding coordinates before rotation can be obtained by the reverse transformation process. Namely, the isochronous grid of the solar path on the street-view panorama (Fig. 8 (a)) can be standardized by street-view projection transformation (see Appendix E).

Each solar trajectory pixel in the rotted panorama stands for a specific time of the sun appearing at the corresponding position in the initial street view. Overlapping the sky segment with solar trajectories, we can obtain the time distribution of sunshine within the street environment (Fig. 8(b)). We mirror the image overlapped with the solar path along the winter solstice edge so as to make the corresponding solar date apparent with the pixels' vertical coordinates (Fig. 8(c)). The equation of time (EQT) calibration Nourian (2020) is adopted to calibrate the hour difference throughout the year (see Appendix F).

Denoting the horizontal axis by the time and the vertical axis by the date, the time distribution of the direct sunlight is illustrated by the intersection of the sky and solar trajectories (Fig. 8(d)). In the same way, we can obtain the time distribution of the reflection from a specific facade. The intersection between the vertical line and overlapped sky indicates the dates when direct sunlight is available at a specific time.

Applying the time distribution estimation approach to the 0–1 matrixes representing solar orientation, we obtain a 0–1 matrix, *KT*, denoting the time distribution of direct sunlight. In the same way, the time distribution of reflection from all visible facades, $u \in U$, is denoted by a set of 0–1 matrixes, OT_u, corresponding with the sample site. Here, u stands for the index of facades. As shown in Fig. 9(a), the row and column index stand for the date and time, respectively. Each nonzero cell in the matrixes indicates the reflection or sunlight scenarios at the moment corresponding with the date–hour coordinates $[d, t]$.

2.3.2. Time specific classification of reflection scenarios

The time-specific reflection number at the sample site is obtained by

$$\mathrm{AT} = \sum_{u \in U} \mathrm{OT}_u, \tag{10}$$

where $u \in U$ is the set of visible GCW surfaces in the panorama. $AT\ (d,\ t)$ stands for the number of reflection apparent in the corresponding date and time of $[d,\ t]$. This paper defines the scene of the sample site receiving sunlight reflected from the facades as the reflection scenario (Fig. 9(b)). The time distribution matrix of the reflection scenarios is represented by a 0–1 matrix, RT, based on the nonzero cells in the matrix AT, as follows:

$$RT(d,\ t) = \begin{cases} 1, & AT(d,\ t) > 0 \\ 0, & AT(d,\ t) = 0 \end{cases},\ \forall t,\ \forall d. \tag{11}$$

In the case that the sample site receives reflected sunlight from several facades at the same time, the scene is defined by the reflection focus scenario in this study. This is a subclass of the reflection scenario. The 0–1 matrix indicating the reflection focus scenario's time distribution is defined by

$$RFT(d,\ t) = \begin{cases} 1, & AT(d,\ t) > 1 \\ 0, & AT(d,\ t) \le 1 \end{cases},\ \forall t,\ \forall \mathrm{d}. \tag{12}$$

The reflection focus scenario is classified as the scene with significant visual interruption to people near the sample site (Fig. 10(a)). The reflection focus scenario is defined to provide indicators for visual impact evaluation.

Overlapped with the time distribution matrix of direct sunlight, the reflection scenarios are divided into two subclasses. The scene when the sample site receives the direct sunlight and reflection at the same time (Fig. 10(b)) is defined as the sunny reflection scenario. The sample site is considered to receive additional and unnecessary heat on hot days. Reflection happening in the scene without direct sunlight (Fig. 10(c) is defined as the shaded reflection scenario. The reflected irradiation is considered to supply the shortage of solar irradiation in cold seasons. The time distribution of these two subclasses can be obtained by a matrix

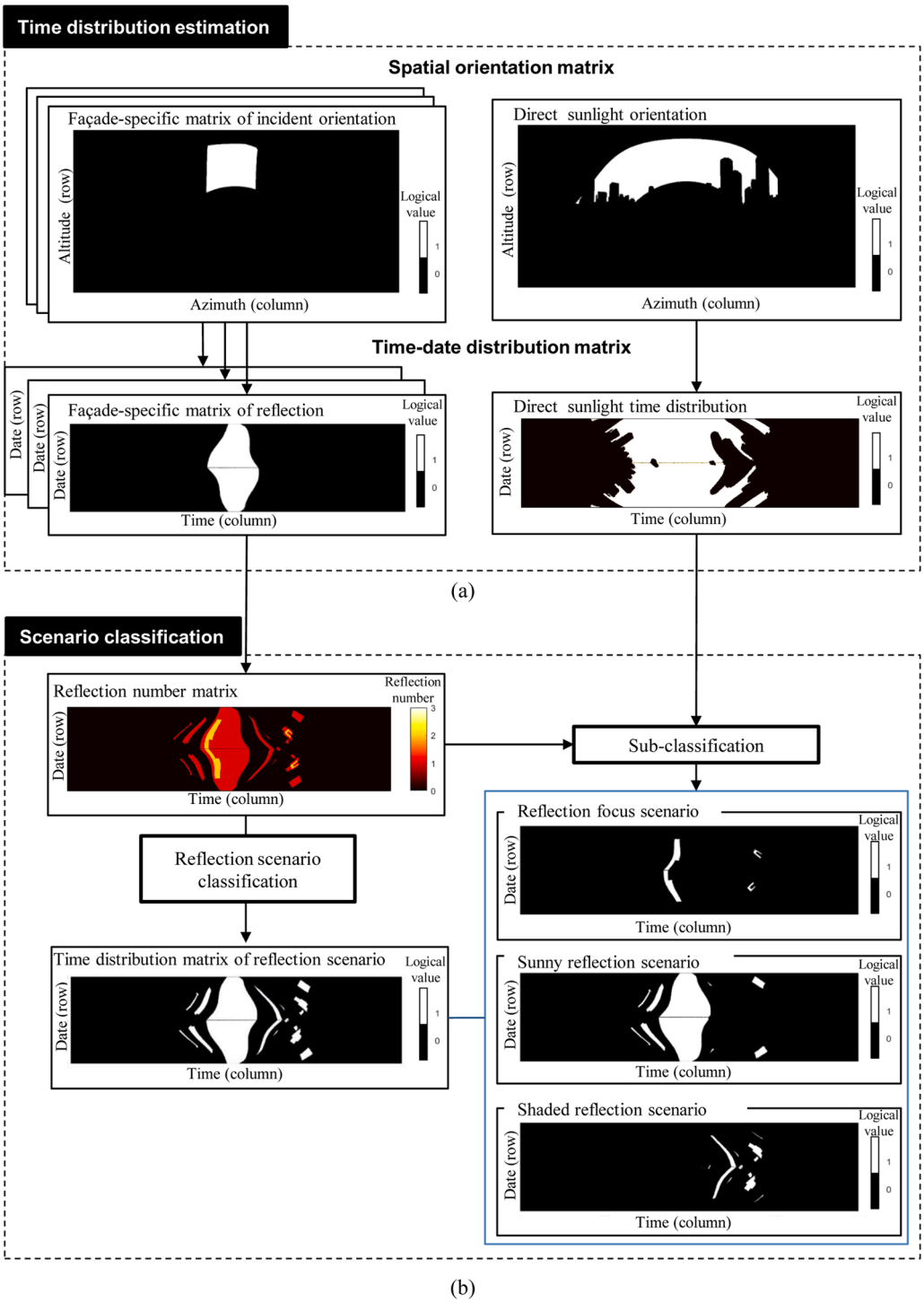

Fig. 9. Estimation approach of time distribution module.

operation as follows:

$$RIT = KT. * RT, \tag{13}$$

$$SIT = RT - RIT, \tag{14}$$

where *RIT* and *SIT* are 0–1 matrixes denoting the scenario classification result by 1 (Fig. 9(b)). These two matrixes are expected to provide the indicators for the reflected sunlight's impact in both negative and positive terms.

2.3.3. *Time-series data of reflected irradiation at the sample site*

Time-series data of reflection are obtained by transforming the incident matrixes input with the reflection parameters (Fig. 11). The time distributions of the azimuth, zenith, and incidence angle are estimated as the matrixes of φ_{dt}, θ_{dt}, and α_{dt}. These data are used for locating the reflection's position on the panorama as well as providing parameters for the reflection intensity estimation.

2.4. *Spatiotemporal distribution module*

This module treats the study area as a network system of sample sites

Fig. 10. Sub-classifications of reflection scenario: (a) reflection focus scenario, (b) sunny reflection scenario, (c) shaded reflection scenario.

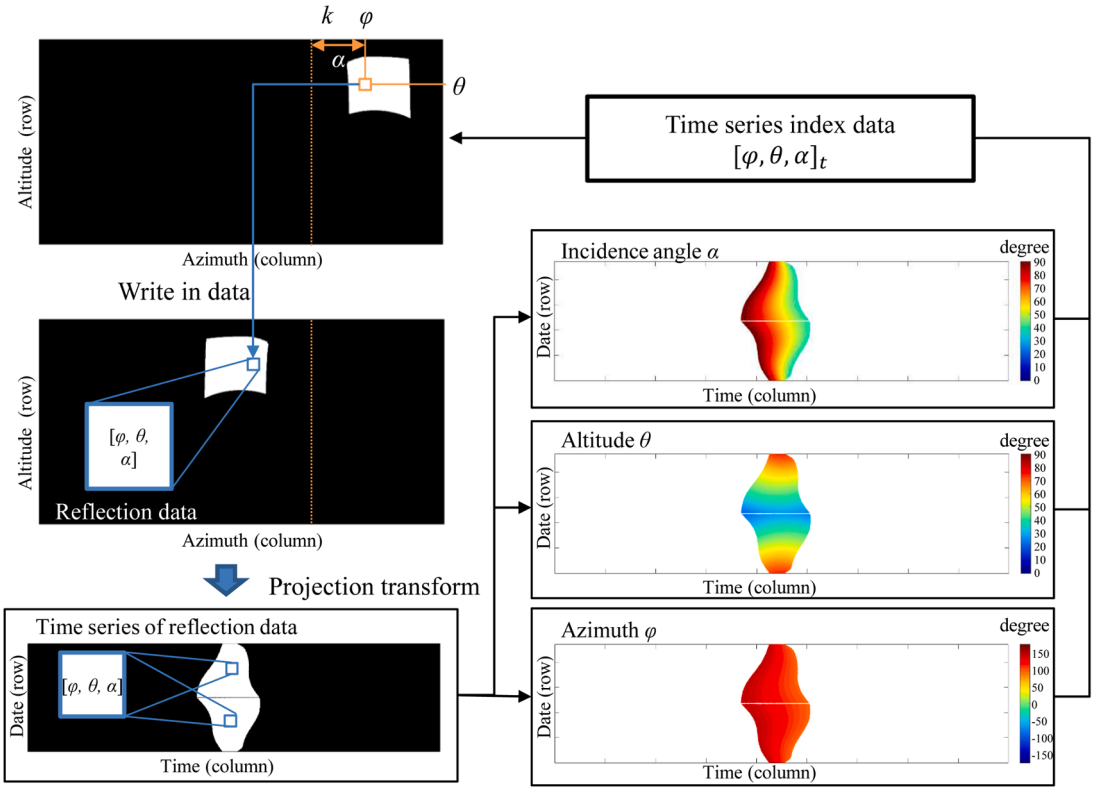

Fig. 11. Estimating time-series data of reflection based on projection transformation.

and aims to estimate the time-specific distribution of reflection scenarios. The module input requires a set of reflection time distribution matrixes estimated based on the panoramas collected at all sample sites.

2.4.1. Time-specific distribution of reflection scenarios in the study area

For a certain reflection scenario, we abstract its time distribution at all sample sites to build a multi-layer 0–1 matrix, $\{RT\}_s$. The layer index, *s,* stands for the sample site. The time-specific impact region of reflection scenarios in the study area is denoted by the time series of the scenario ratio as follows:

$$RTR = \frac{\sum_{s=1}^{m} RT_s}{m} * 100\%, \tag{15}$$

where *m* stands for the total number of the sample sites in the study area, and $RTR(d,t)$ represents the ratio of the reflection scenario at the date-time coordinate [*d, t*]. As shown in Fig. 12, the array of cells in the time direction indicates the daily impact region variation, while the array in the date direction stands for seasonal change of the impact region throughout the year.

The time-specific scenarios of sample sites are visualized by filling the sample sites based on cells with the same date-time coordinates in all facades' time distribution matrixes. Abstracting the spatial distribution in a certain time sequence, we obtain the spatiotemporal distribution of the reflection scenario in the study area.

2.4.2. Accumulative spatial distribution tendency of reflection

For each sample site, the accumulative duration of the reflection scenario at the day and the year level can be obtained by

$$CD_d^{day} = \frac{24}{t_{max}} \sum_{t=1}^{t_{max}} RT\ (d,\ t),\ \forall d, \tag{16}$$

$$CD^{year} = \frac{365}{d_{max}} \sum_{d=1}^{d_{max}} CD_d^{day}, \tag{17}$$

where t_{max} and d_{max} stand for the size of the time distribution matrix in the time and date dimensions, respectively. This study takes the accu-

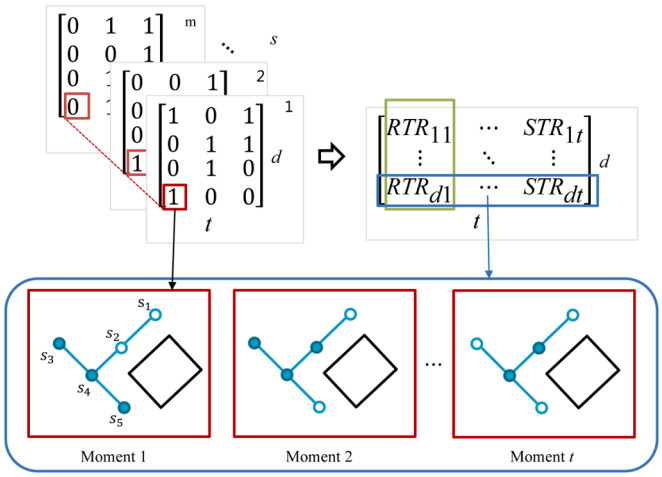

Fig. 12. Visualization of time-specific reflection scenarios.

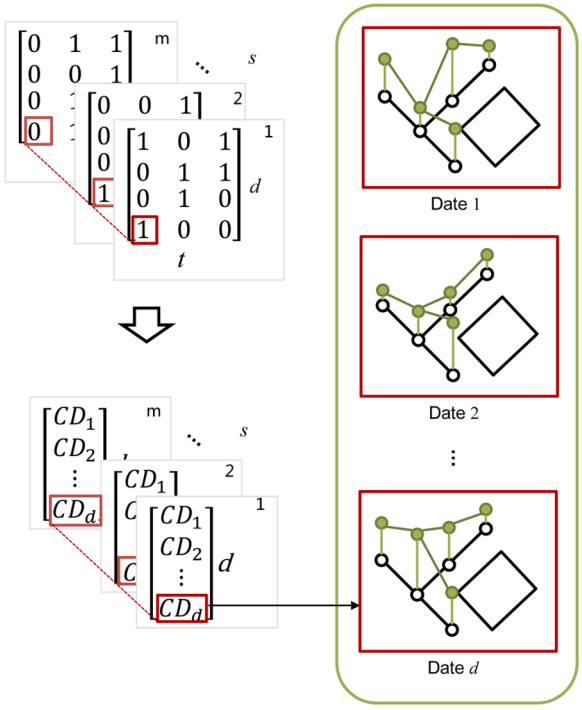

Fig. 13. Visualizing the spatiotemporal distribution tendency of reflected sunlight.

mulative duration of reflected sunlight as the indicator for its spatial distribution tendency at a certain sample site. The accumulative spatial distribution tendency in the study area is obtained based on indicators of all sample sites in the network.

Representing indicators by the height of vertical segments at the sample sites in the network, we can plot the accumulative spatial distribution of reflected irradiation in the study area. The duration variation along streets is visualized by the polygons connected by the end points of neighboring segments in the network (Fig. 13). With the results of a series of time intervals, we can obtain the spatiotemporal distribution of reflected sunlight from GCW facades in the study area.

By the same method, we can obtain the accumulative spatial and spatiotemporal distribution of a subclass reflection scenario based on its time distribution matrixes at the sample sites in the study area.

2.4.3. Time-series intensity of reflected irradiation in the study area

Base on the model of Li and Ratti, 2019, we can calculate the time distribution of direct solar irradiation by

$$F_{dt}^{DIR} = \frac{24I_d^{OBS}}{t_{max}} \frac{KT_{dt} * f_{dir}(\theta_{dt})}{\sum_{t=t_1}^{t_2} f_{dir}\left(\theta_{dt}\right)}, \ \forall t, \ \forall d. \tag{18}$$

where θ_{dt} is the solar zenith at $[d, t]$ obtained in the parameter matrix in the Section 2.3.3, and KT_{dt} is a Boolean value in the sunlight time distribution matrix. t_1 and t_2 stand for the time of sunrise and sunset, respectively, known from solar time distribution *KT*. I_d^{OBS} is the total sunlight irradiation for 1 day, known from the ground station. $f_{dir}(\theta_{dt})$ is the function describing the relation between solar zenith angle and solar radiation intensity. Setting the time interval T by 1 hour, we can estimate the hourly sunlight radiation intensity at the sample site by

$$I_{dT}^{DIR} = I_d^{OBS} \frac{\sum_{t\in T} KD_{dt} * f_{dir}\left(\theta_{dt}\right)}{\sum_{t=t_1}^{t_2} f_{dir}\left(\theta_{dt}\right)}, \tag{19}$$

where $t \in T$ stands for the time interval in 1 h. Accordingly, the reflected irradiation from GCW facades is obtained by

$$F_{dt}^{REF} = \frac{24I_d^{OBS}}{t_{max}} \frac{\sum_{u\in U} g(\gamma_{dt}) f_{dir}(\theta_{dt}) OT_{dtu}}{\sum_{t=t_1}^{t_2} f_{dir}\left(\theta_{dt}\right)}, \tag{20}$$

where $g(\gamma_{dt})$ is the model describing the relation between the incident angle and the reflection ratio, and γ_{dt} is an incident angle between the ray and surface, which is known by

$$\gamma_{dt} = \sin^{-1}(\sin\alpha_{dt}\cos\theta_{dt}). \tag{21}$$

The hourly intensity of sunlight reflection from GCW facades is quantified by

$$I_{dT}^{REF} = \sum_{t\in T} F_{dt}^{REF}. \tag{22}$$

The accumulative intensity at the day level is obtained by

$$I_d^{REF} = \sum_{t=t_1}^{t_2} F_{dt}^{REF}. \tag{23}$$

This study takes the intensity of reflected sunlight as the indicator of its thermal impact on the street-level environment at the sample site. The spatial distribution of reflected sunlight's thermal impact in the study area is obtained based on indicators of all sample sites. The thermal impact's spatial distribution tendency is visualized by the same method mentioned in the Section 2.4.2. With a time series of indicators, we can quantify the spatiotemporal intensity of reflected heat from GCW facades in the study area.

2.5. Model validation

2.5.1. Verification of time distribution module

We verify the accuracy of proposed model through the reflection simulation. As shown in Fig. 14(a), there are 16 sampling points in the surrounding area of the GCW building complex, and the spacing between each sampling point is 10 m. Through panoramic image photography, the street view panorama at each sampling point is collected on a sunny day.

We applied this method to calculate the accumulative daily reflected irradiation time for all observation points in the data collection area. We estimated the daily duration on the 22nd of each month, and compared the estimated results with the 3D simulation results (Ecotect Analysis v2011.0.0.2). As shown in Fig. 14(b), the estimated result and the model simulation result have a good degree of fit. The difference between the estimated and simulated results is due to the position and size of 3D models used in the simulation, which did not fix the actual condition.

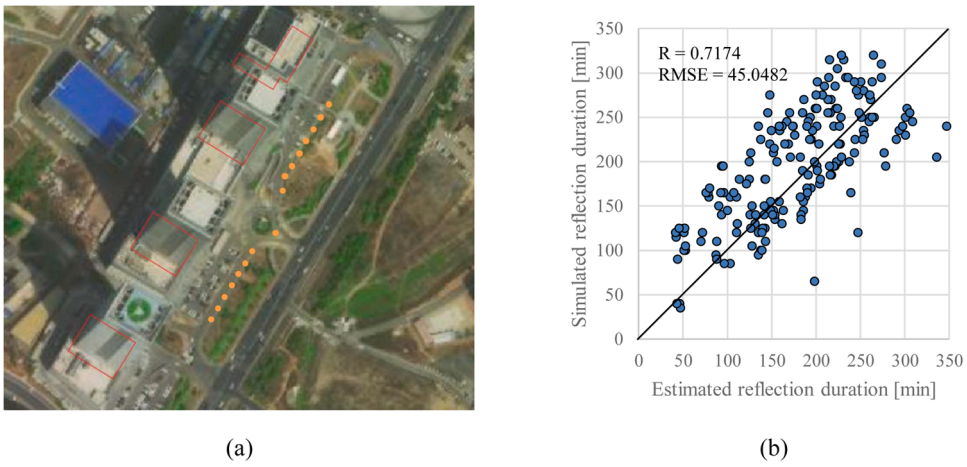

Fig. 14. Model validation using reflection simulation: (a) Sampling points and building footprint map; (b) Fitness between estimated and simulated result.

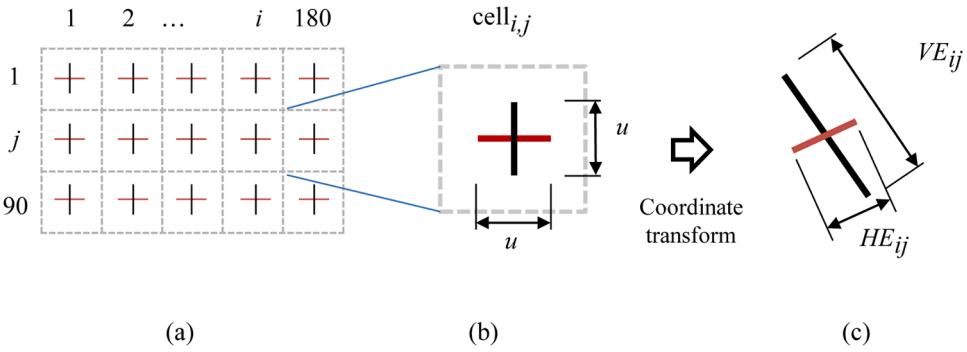

Fig. 15. Estimation of projection transformation error: (a) simple grid, (b) errors before transformation, (c) errors after transformation.

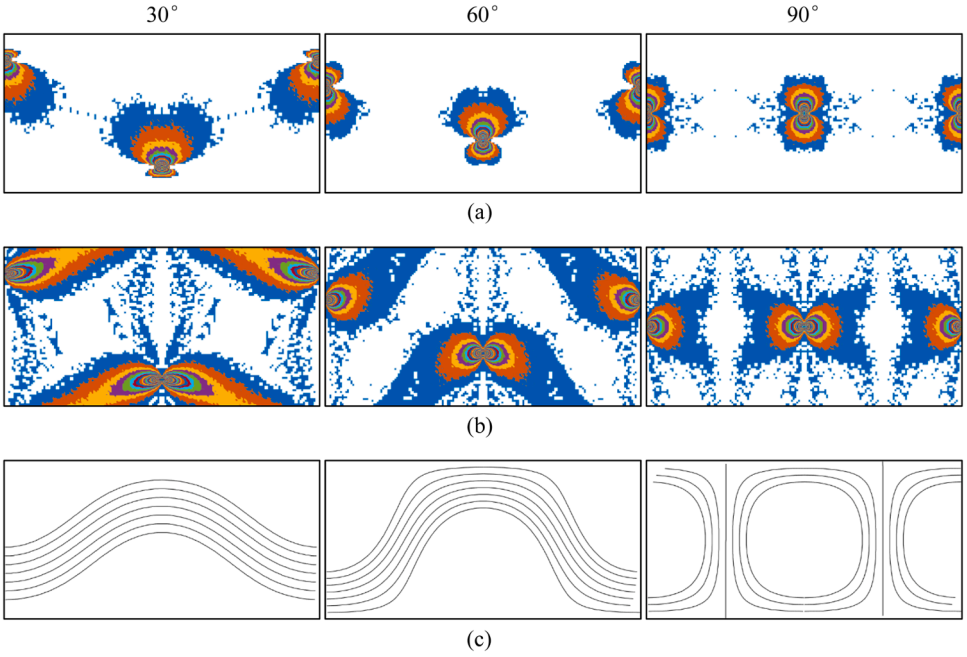

Fig. 16. Error estimation result of panoramic projection transformation: (a) horizontal error, (b) vertical error, (c) Solar trajectories.

We will validate our method based on the observed data in the future.

2.5.2. Error verification of coordinate transformation

Panoramic projection transformation not only changes the objects' coordinates in an image, but also the size of the region. The errors,

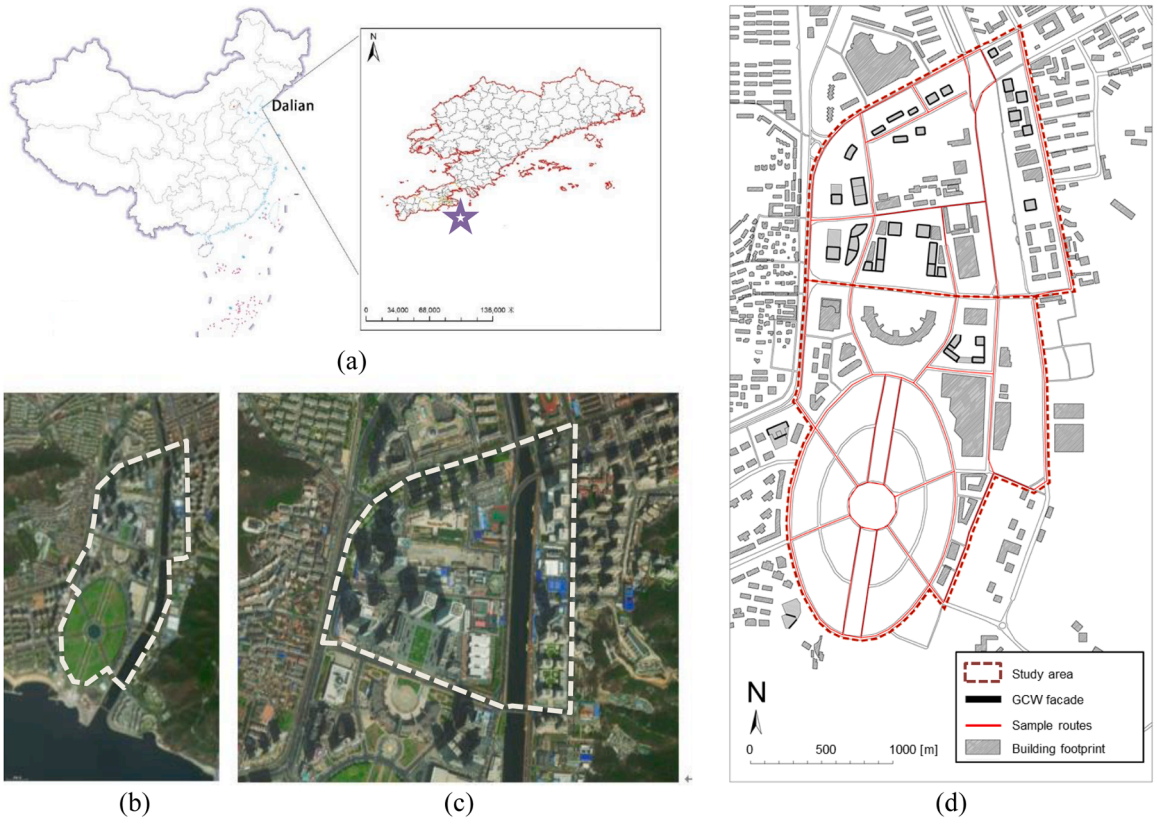

Fig. 17. Study area and data: (a) location of Dalian, (b) Xinghai Square region, (c) Xinghai commercial district, (d) Street network and building footprint layout.

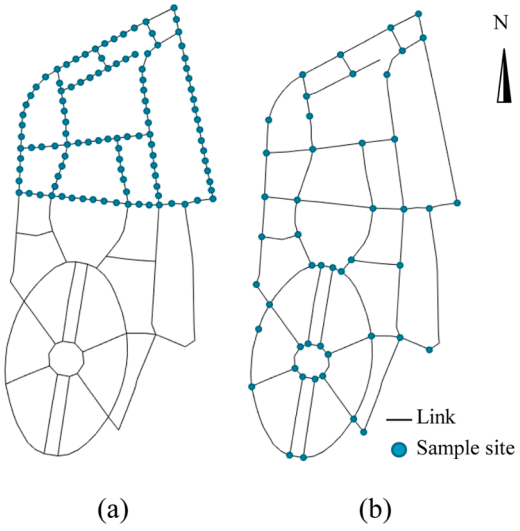

Fig. 18. Spatial locations of sample sites in the street network: (a) the local region, (b) the wide region.

which are distributed homogeneously in the original image, were enlarged/reduced accordingly. The error size reduction in the transformed map could be ignored. The enlargement of errors owing to image calibration and projection coordinate transformation was examined in this study.

Taking the 900 × 1800-pixel panorama as an example, this study divided the image area into grids with size-10 pixels (2° spherical system). The resizing of horizontal and vertical errors was examined by two segments (5 pixels corresponding to 1°) crossing at the center of the grid (Fig. 15(a)). The error after transformation was obtained by the coordinate transformation of the segment's end points. The error magnification ratio of projection coordinate transformation was estimated by

$$HR_{ij} = HE_{ij}/u, \tag{24}$$

$$VR_{ij} = VE_{ij}/u, \tag{25}$$

where HE_{ij} and VE_{ij} are the length of the horizontal and vertical segments, respectively, after projection transformation, and u is the error size in the original image.

The front direction calibration only shifts the image horizontally within the same size, so it would not change the size of objects and the error. The mirroring approach, which reverses the horizontal sequence of pixels in a certain region, also had no influence on region size. Error magnification was estimated by pitch rotations with the angles of 30°, 60°, and 90° The estimation result is shown in Fig. 16. The color white in the figure indicates the area with error reduction. The blue area has an error magnification ratio of between 1 and 2. Thus, the potential reflection source area is mostly located in the area with error reduction, indicating that the projection transformation does not have a significant impact on the accuracy of the measurement.

3. Case study

3.1. Study area and available data

This study selected the Xinghai business district in the city of Dalian (N38.9°, E121.6°) as the study object. Dalian (N38.9°, E121.6°) is a coastal city located on the southernmost side of Liaodong Peninsula, China. The Xinghai business district is located on the northern side of Xinghai Square in the Shahekou district. As shown in Fig. 17(b), the study area contains the business district, square, and surrounding blocks. The area is 1.46 km^2 and contains five urban arterial roads and

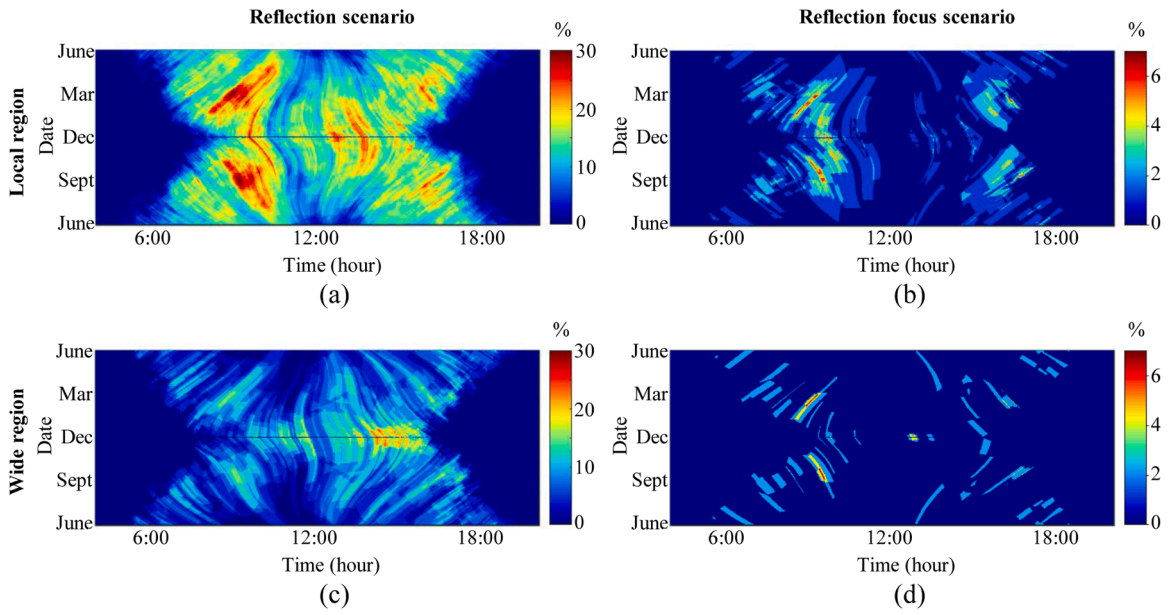

Fig. 19. Time-distribution frequency of reflected sunlight received in study area.

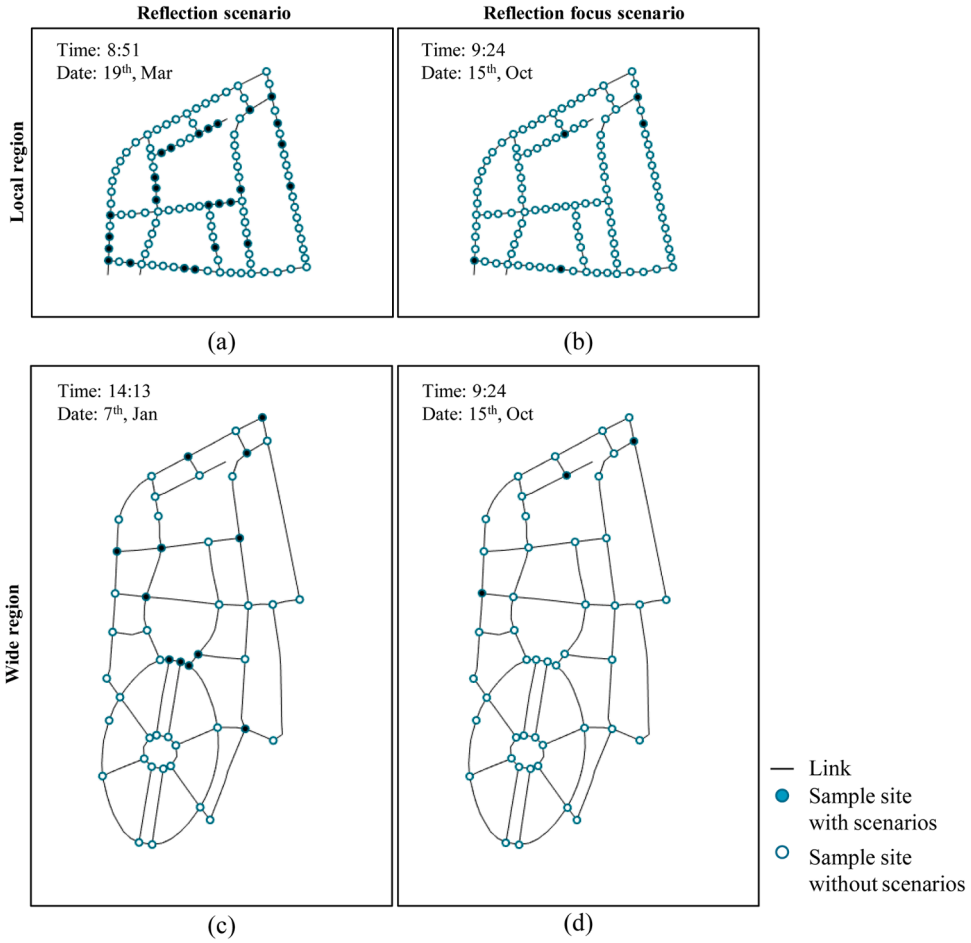

Fig. 20. Scene detection of the most impacted region of reflected sunlight: (a) reflection scenario in the local region, (b) focus scenario in the local region, (c) reflection scenario in the wide region, (d) focus scenario in the wide region.

five secondary roads. There are 45 high-rise buildings in the study area, and most of the GCW buildings were developed after 2004 (Fig. 17(c)). The data of street network and GCW facades were collected and indexed based on the OpenStreetView map (Fig. 17(d)).

Panorama data were collected on July 27th, 2021, which was a sunny day with few clouds. The collection was done at noon

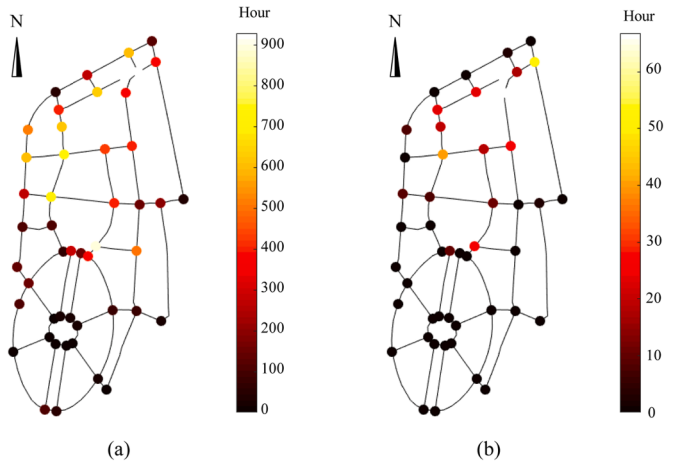

Fig. 21. Annual accumulative duration of reflected sunlight at road intersections: (a) reflection scenarios, (b) reflection focus scenario.

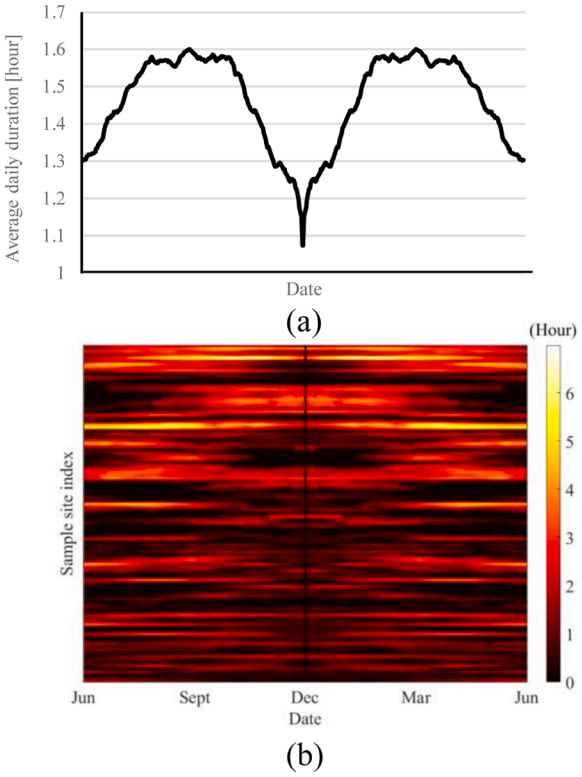

Fig. 22. Variation of reflected sunlight at different sample sites throughout the year: (a) average daily duration of all sample sites (b) site-specific daily duration.

(12:00–13:30) to decrease the interruption of shading between buildings. Based on the street map, we selected the sample routes shown in Fig. 17(d) to collect street-view data.

Sample sites within the commercial district were chosen to study the reflected sunlight's influence on the local region (107 sample sites within an interval of 50 m, shown in Fig. 18(a)). To examine the reflection sunlight's visual impact to traffic safety, we chose the sample sites at the road intersections (48 sample sites in Fig. 18(b)) to conduct a wide-region analysis.

3.2. Time-specific reflection scenario detection

3.2.1. Time distribution of scenario ratio

Applying the proposed method, this study first estimated the time-specific ratio of reflection scenarios in the local and the wide region. Fig. 19 shows the time distribution of reflection scenarios within local and wide regions. The position of each cell represents a specific time and date. As shown, sunlight reflection is common in the local region. Around 30% of sites have sunlight reflection at the same time in the worst case.

Moreover, local and wide regions show different distribution tendencies. For the sites in the local region, continuous sunlight reflection has a high tendency in the mornings of spring and autumn (dark red cells in Fig. 19(a)). For the intersections in the wide region, continuous reflection has a high tendency in the winter afternoons (red cells in Fig. 19(c)).

Fig. 19(b)–(d) shows the time distribution of reflection focus scenario in the local and the wide region. It indicates a risky condition, which may lead to significant nuisance to pedestrians and drivers. The top of values of Fig. 19(b)–(d) are smaller. This means that reflection focus seldom happens at several sites at the same time. Comparison between Fig. 19(a) and (b) shows that the reflection focus scenario is a temporal condition with less continuity in the time sequence.

3.2.2. Time-specific reflection scenarios in the study area

Taking the date with the largest scenario ratio as the examples, this study abstracted the most impacted region of reflection in the local and wide region. As shown in Fig. 20, the distribution of the reflection scenarios has a certain spatial continuity in the selected case. The result of the local region (Fig. 20(a)) shows that pedestrians will continue to receive the glare nuisance within the range of 100–200 m. The result for the wide region shows that drivers may be continuously disturbed by reflected glare at several intersections (Fig. 20(c)).

The reflection focus results (Fig. 20(b)–(d)) show that although the reflection focus scenario has a small impact range, in the case of reflection focus distributed on the same road, it may cause moving pedestrians or drivers to continuously suffer from glare nuisance, which come with a high risk of safety hazards (Fig. 20(b)). At the same time, the reflection focus that occurs at arterial roads' intersections may affect multiple drivers at the same time, increasing the risk of traffic accidents.

3.3. Quantification of spatial distribution tendency

3.3.1. Accumulative reflection duration throughout the year

This study first estimated the annual accumulative duration of the reflection scenario and reflection focus scenario at each road intersection in the wide region. As shown in Fig. 21(a), the maximum accumulative duration of the reflection scenario extends 900 h at the year level. The reflection focus only occupies a small part of the reflection scene, but the annual accumulative duration at some sample sites still reaches 60 h (Fig. 21(b)). The result of the reflection scenario at each road intersection shows a decreasing trend as the distance from the high-rise area increases. The distribution of the reflection focus scenario shows some difference. The values in the northwest sites with a wide field of vision are significantly higher.

3.3.2. Daily accumulative estimation result

As shown in Fig. 22(a), the daily average accumulative durations (the mean value of all sampling sites) of reflection scenarios in the local region vary between 1 and 1.6 h. They peak in the spring and autumn, and become lower in the summer and winter. Fig. 22(b) shows that the variation of a single sampling site has a degree of continuity in the date dimension. The range of daily accumulative duration varies by up to 7 h (Fig. 22(b)). Therefore, it is difficult to describe the characteristics of dynamic variation throughout the year by the average values in the temporal and spatial dimensions.

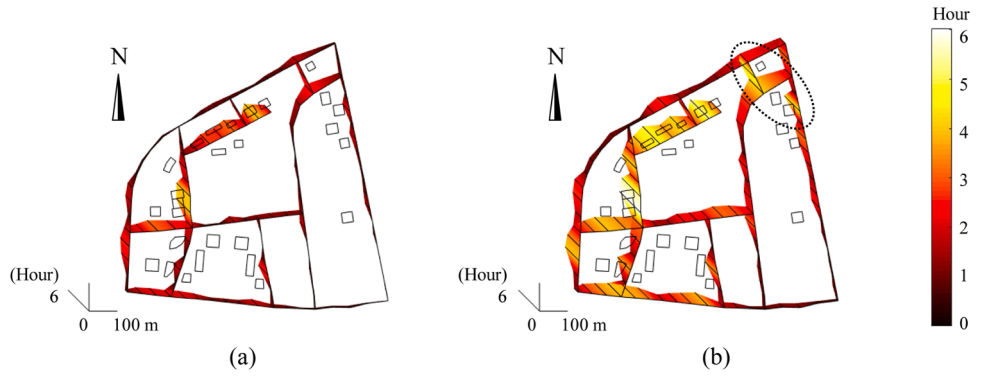

Fig. 23. Spatial distribution tendency of daily reflection scenario duration in the local region: (a) mean value of daily duration, (b) peak value of daily duration.

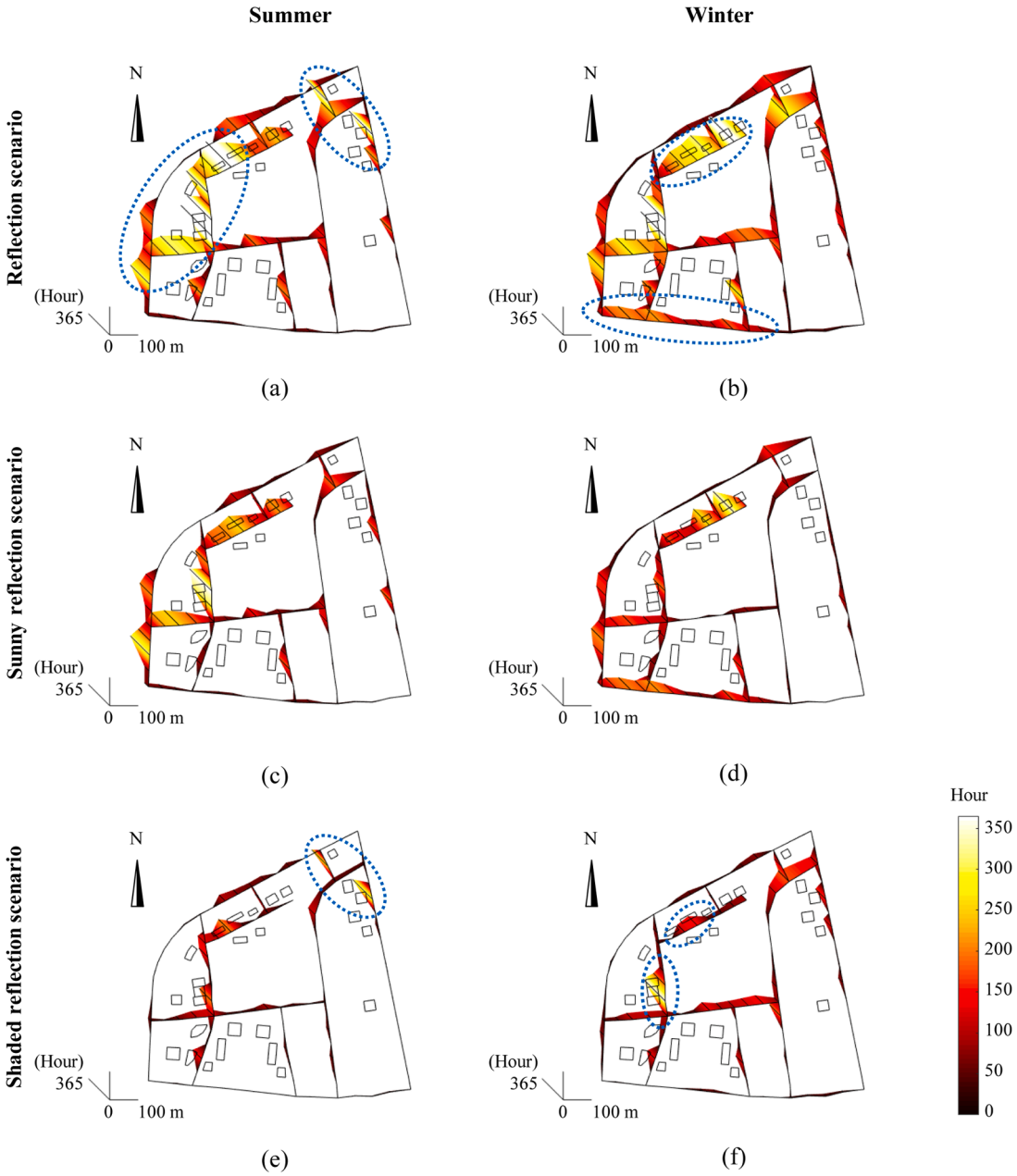

Fig. 24. Seasonal accumulative duration of reflected sunlight: (a) & (b) reflection scenario in summer and winter, (c) & (d) sunny reflection scenarios, (e) & (f) Shaded reflection scenarios.

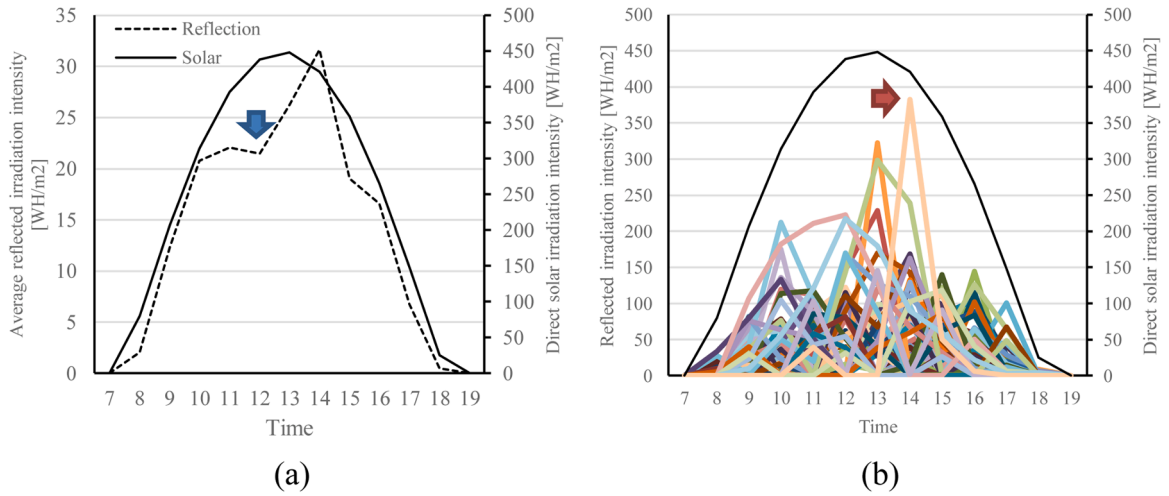

Fig. 25. Correlation between direct solar radiation and reflected irradiation: (a) regional global result, (b) site-specific result.

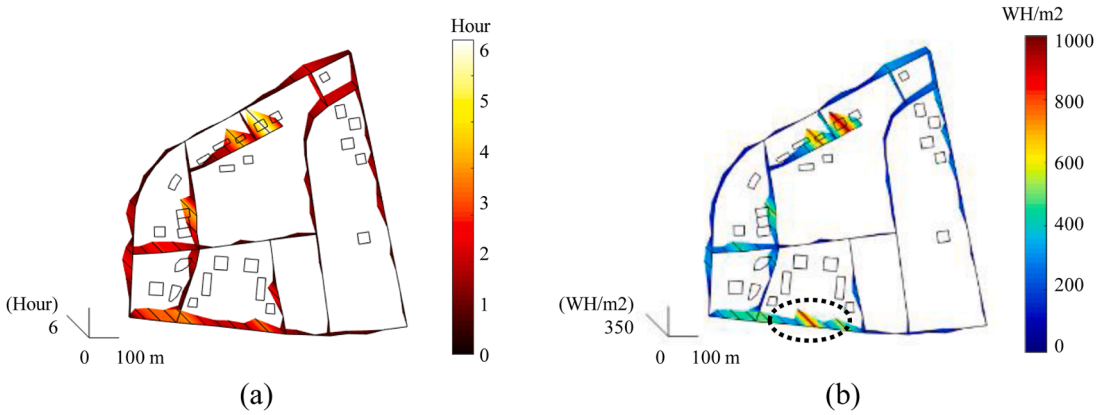

Fig. 26. Visualization of daily accumulative reflection impact: (a) scenario duration, (b) reflected irradiation intensity.

This paper uses the peak values of the daily duration to indicate the maximum spatial distribution tendency of reflected sunlight at each sample site at the day level. Comparison between the peak values (Fig. 23(b)) and mean values (Fig. 23(a)) of daily duration shows that there are higher daily duration peaks in the northeastern part of the study area (dash circle in Fig. 23(b)), and the mean values of these parts are not significant. The date-specific duration estimation allows us to discover the worst conditions from huge, complex spatiotemporal scenarios, and to make more accurate assessments of the impact of reflected sunlight.

3.3.3. Seasonal accumulative distribution result

Fig. 24 shows the seasonal accumulative duration of reflected sunlight in the local region. The whole year is divided into four periods, taking the solstice or equinox day as the middle date, and each period includes around 91 days. The duration of the reflection scenario, sunny reflection scenario, and shaded reflection scenarios in different periods were separately analyzed.

There is a significant difference in the spatial distribution of reflected sunlight in the winter and the summer periods. In the summer, the reflected sunlight mostly occurs in the secondary street areas with dense high-rise buildings (the dash circles in Fig. 24(a)), while more reflection occurs in the open areas (the dash circles in Fig. 24(b)) in the winter. The overlap analysis with sunlight reveals that the spatial distribution of sunny reflection scenario is similar to the reflection scenarios, and it more accurately represents the additional thermal effect of reflected sunlight in the study area (Fig. 24(c)). The shaded reflection scenario indicates the reflected sunlight that reached the shadow area. Although some areas have a longer reflection duration (the dash circles in Fig. 24(e)), they do not necessarily receive a significant heat load from the reflective facades. The results in the winter period show that the secondary road area in the shadow of high-density high-rise buildings receives the reflected heat from the GCW (the dash circles in Fig. 24(f)), which has a promotional effect on alleviating road icing in cold winters.

3.4. Spatiotemporal intensity of reflected irradiation

3.4.1. Spatiotemporal intensity at the day level

Based on the model of Li and Ratti, 2019, we assumed that the direct solar irradiation intensity and the altitude angle satisfied a sine function relationship. Taking the longest duration day as an example, this study estimated the hourly solar irradiance in the unobstructed condition based on the total direct solar irradiation (3.1 kWh/m^2, November 7th, 2020) from the ground stations. We assumed that the incident angle and the reflectance of the glass curtain walls in the study area satisfied the simulation results of Danks et al. (2016).

We applied the proposed model to calculate the reflection intensity of all sample sites in the study area. As shown in Fig. 25(a), there is a positive correlation between the solar irradiation intensity and the average reflected irradiation intensity (mean value of all sample sites). It is found that the reflected radiation intensity will temporarily decrease at noon (the blue arrow in Fig. 25(a)). The possible reason is that the light direction is parallel to the east-west facades at noon, which may decrease the possibility of reflected sunlight reaching the sample sites.

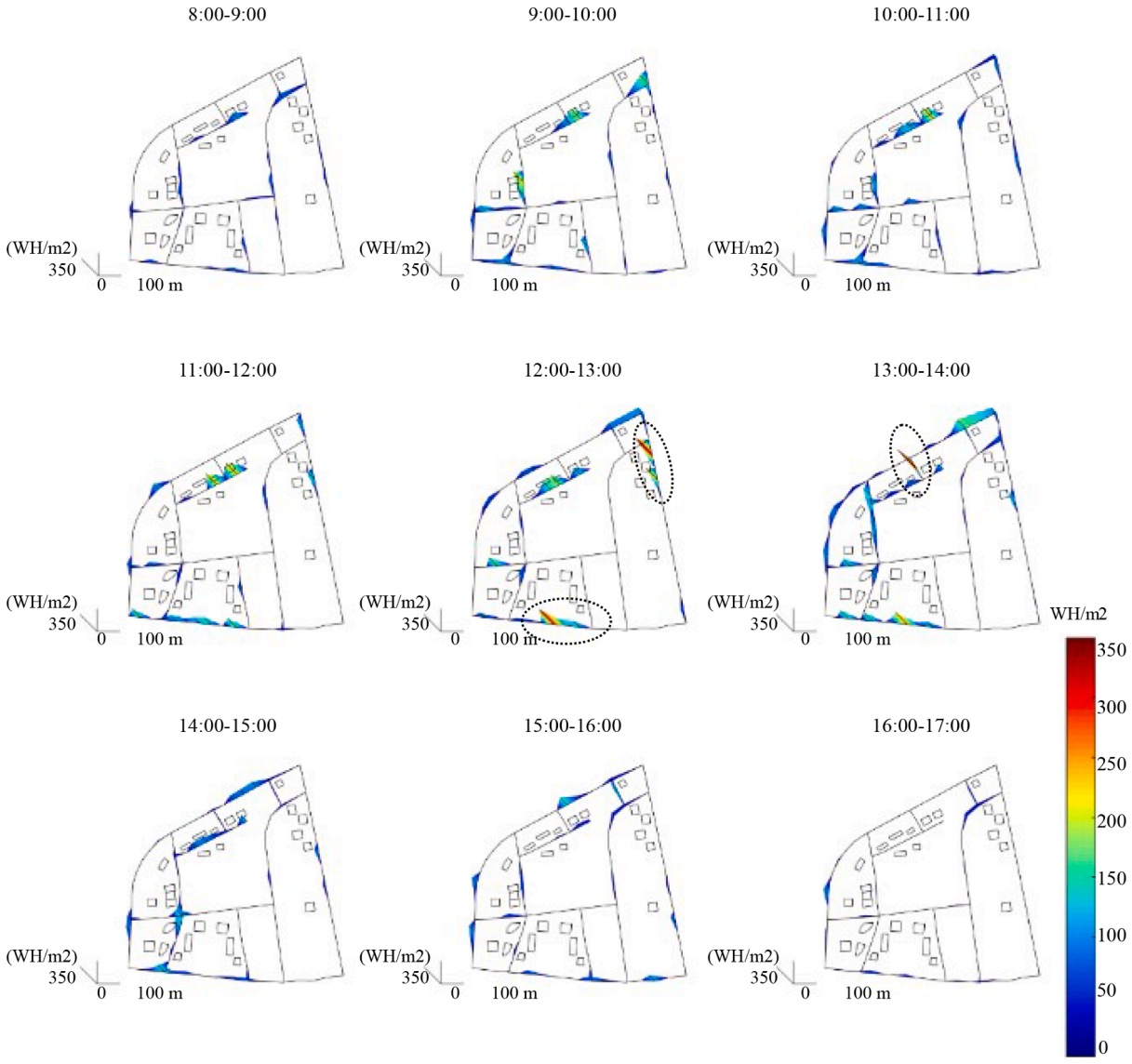

Fig. 27 h. Hour-specific spatial distribution of reflected irradiation intensity.

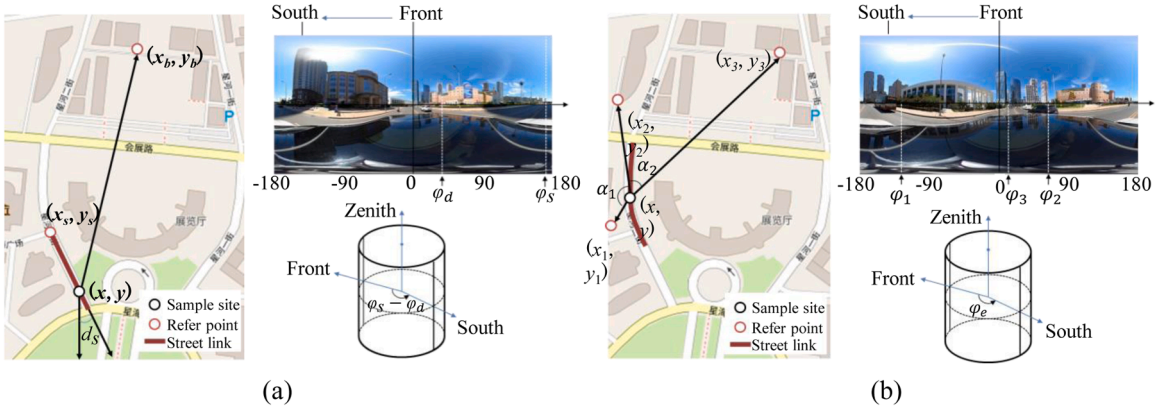

Fig. B.1. Front orientation calibration approach: (a) site on straight street, (b) site on non-linear street.

In view of the intensity variation at each sample site (Fig. 25(b)), the peak intensities at sample sites are significantly higher than the average level of the study area. The reflected irradiation intensity at a specific site reaches 85% (the red arrow in Fig. 25(b)) of solar irradiation. There are significant differences in the irradiation intensity of different locations in the same time interval. However, the peak values of different locations appear at different hours. Therefore, we adopted the time-specific distribution analysis to study the temporal-spatial

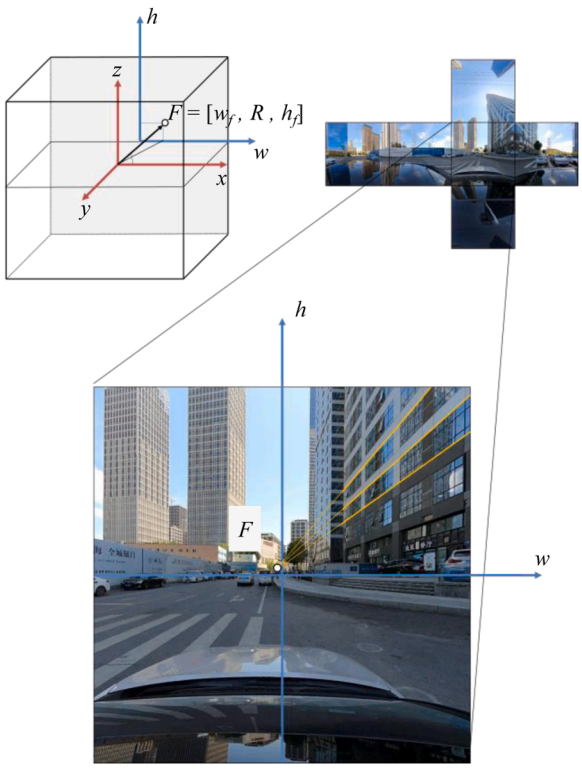

Fig. B.2. Zenith direction calibration approach.

heterogeneity of the reflected irradiation.

3.4.2. Time-specific variation of reflected heat from GCW facades

We estimated the daily reflection duration and reflected irradiation intensity of each sample site in the study area and plotted their spatial distributions (Fig. 26). Through comparative analysis, it can be seen that there is a significant spatial correlation between these two indicators. Long reflection duration is considered as the main reason for the high irradiation intensity. Moreover, although the reflection duration at the middle of the southern main road (dashed circle in Fig. 26(b)) is relatively short, its daily reflected irradiation intensity also reaches 800 WH/m^2. This indicates that the intensity of reflected irradiation is also affected by other factors, such as incidence and solar altitude.

We estimated the hour-specific intensities of reflected irradiation at all sample sites in the study area and visualized their spatial distribution tendency (Fig. 27). With the migration of time, the high-intensity regions change significantly, and the variation does not show continuity in the time and the space dimensions.

Take the periods between 12:00 and 14:00 as examples, the three high-intensity sites (dash circles in Fig. 27) differ greatly from their surrounding sites. However, the high-intensity irradiation temporally appears in a certain hour, and decreases rapidly in the next hour. These results show that the reflected irradiation intensity in the areas with dense high-rise buildings has significant dynamic characteristics, which are difficult to describe by a single scene or accumulative results. Spatiotemporal distribution estimation provides an effective tool for detecting and visualizing the dynamic impact of reflected sunlight.

4. Discussion

4.1. Contribution

This study provided a framework for estimating the spatiotemporal distribution of reflected sunlight based on street-view panoramic data. This method is expected to support risk detection, quantitative assessment, and spatiotemporal characteristics visualization of reflected sunlight impact analysis.

This model classifies the environmental effects of reflected sunlight through the definition of sub-reflection scenarios, and detects spatiotemporal scenes that have significant visual and thermal effects. These data will help city managers take measures and provide early warnings to avoid accidents. The estimated data have higher time and space accuracy, which is convenient for managers to take effective measures with lower cost. The peak reflection duration identification enables us to grasp the common and the most disadvantageous degree of the reflection's impact on the urban space. It is expected to help city government and managers implement policies to reduce the impact of reflection.

This model quantifies the spatial and temporal distribution trend of the dynamic phenomenon of reflected sunlight through the definition of scenario ratio and accumulative duration. The quantification will provide indicators for urban spatial analysis and street environment assessment (e.g., walkability scores). It also enables further study on the mechanism of the street microenvironment and urban residents' activities based on estimated data. At the same time, the acquisition of time-series spatial reflection parameters provides data support for modeling and simulation of the optical and thermal environment of the urban street space. The quantitative work also provides a reference for decision makers to make more effective policies to mitigate the UHI effect while reducing the negative effects of reflective materials.

The visualization of the spatiotemporal distribution will intuitively help planners, urban designers, and landscape designers obtain the dynamic distribution of reflected sunlight so as to reasonably arrange the function and form of the urban street space, for example, by planning indoor pedestrian facilities (e.g., underground passages) on routes with a long reflection duration, avoiding parking lots or outdoor resting spaces in areas with high reflection intensity, and avoid setting up car lanes in areas with high potential for reflection focusing. This model will

(a) (b) (c)

Fig. B.3. Types of glass curtain walls in study area.

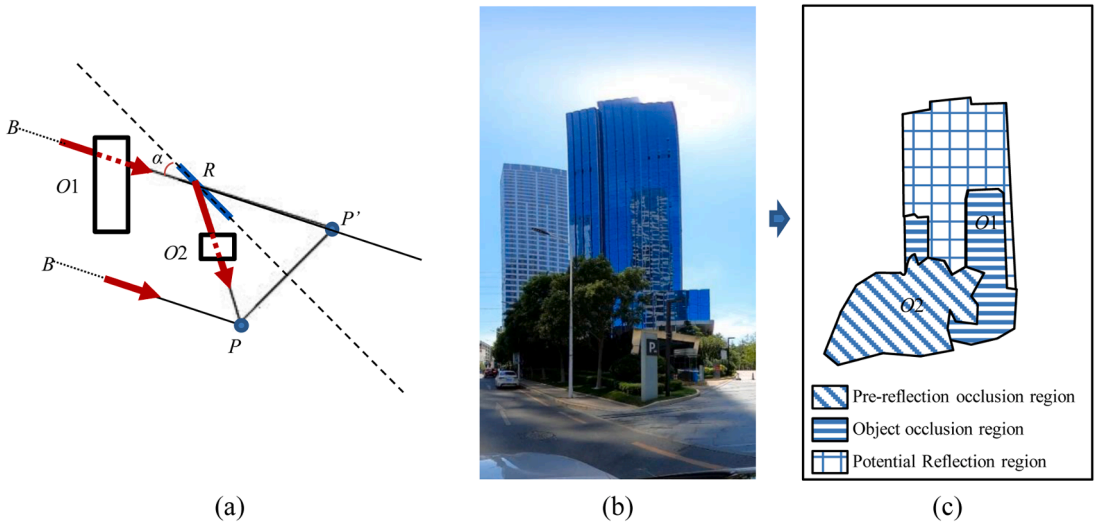

Fig. B.4. Classification of occlusion and potential reflection region in the street view: (a) types of object occlusion, (b) images in the street view, (c) classification result.

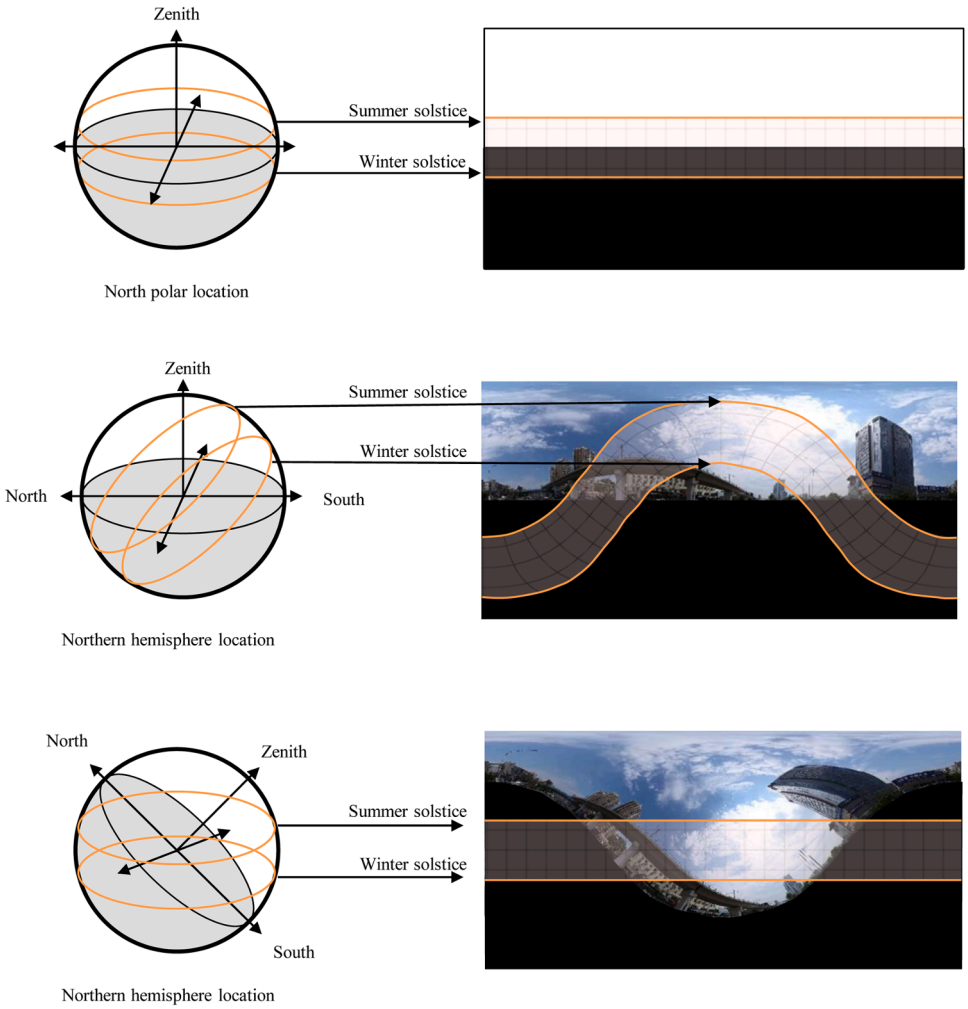

Fig. E.1. Solar trajectories on the panorama at polar location.

help landscape designers set shading facilities and street trees according to the time-specific distribution of reflected sunlight in the street space. It will also help them plant deciduous trees and take anti-slip measures in under-irradiated areas in winter.

The street-view data used in this method can be conveniently obtained through an open-source API or panoramic photography. The calibrating and positing method was also proposed for the condition without open-source data. The proposed method does not rely on the 3D

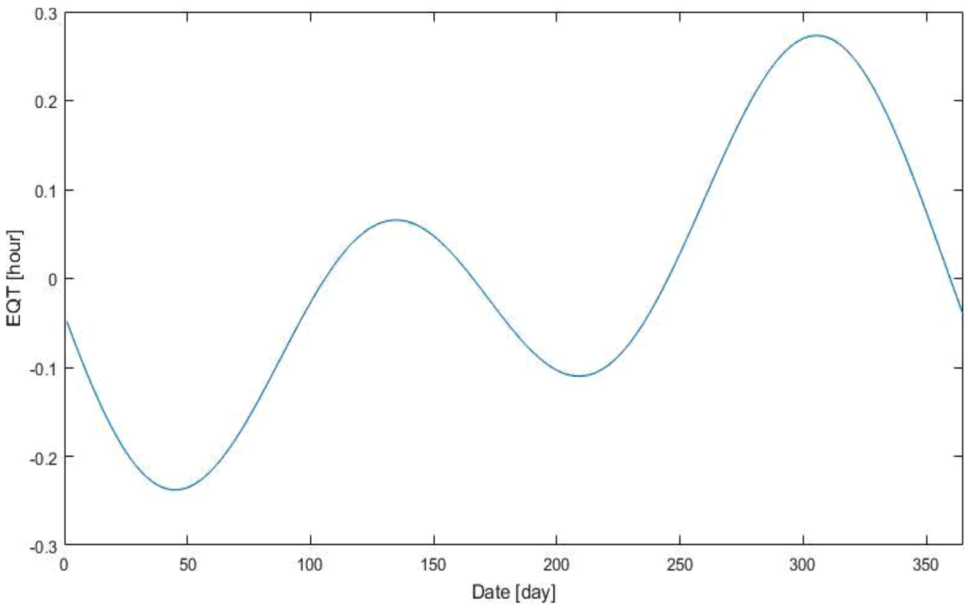

Fig. F.1. EQT value in the year.

building model or detailed urban spatial data, but rather easily purchased data, such as data on the facade orientation and the road map, which can be applied in regions lacking urban information data.

4.2. Limitations and future work

Street-view data may provide big data for analyzing and measuring the urban environment on a wide scale, but they also pose challenges in huge data processing. The analysis scale in this study was limited to the regional scale owing to the time-consuming process of segmentation. Machine learning and image processing are other possible batch processing methods. Image processing has been applied in detecting sky regions in city ranges in previous studies. However, it is still challenging to accurately separate the reflected and direct sky regions owing to the similarities in color. Machine learning has been widely used in the segmentation of the street environment. Specific training sets have also been developed to detect GCW facades in the street view Ying (2021). However, extraction of the reflected sky region in a curtain wall façade is still not satisfying in the current approach. Therefore, the main aim of the study was to construct an estimation model and show the potential of spatiotemporal distribution data in evaluation and planning assistance. Further efforts will be made to develop batch process algorithms in the future.

The quantification proposed in this manuscript is mainly focused on the thermal impact of reflected sunlight. In the street environment, a low-altitude reflection source is easier for pedestrians to perceive and causes more discomfort. The time-series parameters of reflection allow us to estimate the dynamic spatial distributions of reflected glares in people's view field. The quantification of the visual impact on drivers and pedestrians will be studied in the future.

5. Conclusions

This study estimated the spatiotemporal distribution of sunlight and sunlight reflection surrounding the Xinghai business district by proposing a novel method based on the street-view panorama. We obtain the source direction of solar radiation leading to sunlight reflection by image processing. The time distribution of sunlight reflection on selected facades at road intersections and street sites is calculated by panoramic projection transformation. Through the quantification of reflection duration and density, the spatiotemporal impacts of reflected sunlight at street-level are visualized. Based on the results of case study, the conclusions were drawn:

(1) Sunlight reflection is dynamically distributed in the street environment with the variation of time and date. As a sub-category, reflection focus scenarios have a lower probability and different distribution tendency.
(2) The spatial distribution of reflected sunlight at the street level is affected by factors including the urban morphology, street spatial characteristics, and facade orientations.
(3) The spatial distribution of the reflection scenario has a significant seasonal difference. The spatial inconsistency between solar radiation and reflected sunlight makes reflected irradiation play positive or negative thermal effects in different seasons and climatic conditions.
(4) The hourly variation of the total reflected irradiance in the surrounding area of high-rise buildings is significantly related to the solar altitude angle. It is also related to other reflection parameters varying according to time.
(5) The time-specific distribution of the reflected irradiation intensity has significant temporal and spatial heterogeneity, and it is not suitable to be described by a single scene or the accumulative results.

The proposed model can provide data support and reference for urban thermal environment analysis, urban planning, space design, and urban management.

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Declaration of Competing Interest

None

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Appendix A. Notation

Variables & symbols	Description	Unit
GCW	Glass curtain wall	Abbreviation
β	Geographic latitude of view point	Rad.
φ	Azimuth angle of	Rad.
θ	Zenith angle	Rad.
w	Pixel's width coordinate	$w \in [1,\ w_{max}]$
h	Pixel's height coordinate	$h \in [1,\ h_{max}]$
SR	Solar path region	$h_{max} * w_{max}$
DR....	Direct sunlight orientation region	$h_{max} * w_{max}$, 0–1 matrix
KR	Sky segmentation region	$h_{max} * w_{max}$, 0–1 matrix
k	Facade surface direction	Rad.
α	Angle between incident ray and surface on the plane	Rad.
γ	Angle between incident ray and surface	Rad.
$u \in U$	Reflective facades index	Int.
GR_u	reflective region on the surface	$h_{max} * w_{max}$, 0–1 matrix
MR_u	Mirroring of reflective surface region	$h_{max} * w_{max}$, 0–1 matrix
OR_u	Reflection source region, the region of incident orientation causing reflection	$h_{max} * w_{max}$, 0–1 matrix
t	Time coordinate	$s \in [1,\ s_{max}]$
d	Date coordinate	$d \in [1,\ d_{max}]$
KT	Time distribution of direct sunlight	$d_{max} * t_{max}$, 0–1 matrix
AT	Time specific reflection number	$d_{max} * t_{max}$, matrix
OT_u	Time distribution of reflected irradiation from facade *u*	$d_{max} * t_{max}$, 0–1 matrix
RT	Time distribution of reflection scenario	$d_{max} * t_{max}$, 0–1 matrix
RFT	Time distribution of reflection focus scenario	$d_{max} * t_{max}$, 0–1 matrix
RIT	Time distribution of sunny reflection scenario	$d_{max} * t_{max}$, 0–1 matrix
SIT	Time distribution of shaded reflection scenario	$d_{max} * t_{max}$, 0–1 matrix
s	Sample site index	Int.
RTR	Time distribution of reflection scenario ratio	$d_{max} * t_{max}$, matrix
CD_d^{day}	Daily accumulative duration of reflection scenario	Hour
CD^{year}	Accumulative reflection scenario duration at the year level	Hour
F_{dt}^{DIR}	Direct solar irradiance at a specific date and time	W/m^2
I_d^{OBS}	Observed total sunlight irradiation intensity for 1 day	WH/m^2
F_{dt}^{REF}	Reflected irradiance at a specific date and time	W/m^2
I_{dT}^{REF}	Hourly reflected irradiation intensity	WH/m^2
I_d^{REF}	Irradiation intensity of reflected sunlight at day level	WH/m^2
HR_{ij}	Error magnification ratio of the horizontal segments	
VR_{ij}	Error magnification ratio of the vertical segments	
[*x*, *y*, *z*]	Pixel's vector in 3-dimensional Cartesian coordinate system	

Appendix B. Image positioning and calibration

For sample sites along linear street, the orientation error, φ_e, could be obtained by

$$\varphi_e = \varphi_s - \varphi_d, \tag{b1}$$

where φ_s is the azimuth street direction on the panorama, and φ_d stands for the street's actual direction on the street map (Fig. B. 1(a)). Therefore, the coordinates (*x*, *y*) on the street map could be obtained by solving

$$\begin{cases} \tan\varphi_d(x - x_s) = y - x_s \\ \tan(\varphi_b - \varphi_e)(x - x_r) = y - y_b \end{cases}, \tag{b2}$$

where (x_s, y_s) are the end point coordinates of the street link, (x_b, y_b) are referred objects (e.g., landmark towers) with known coordinates, and φ_b is the reference object's azimuth on the panorama.

For the site along the non-linear street, the coordinates and orientation errors were calibrated by the azimuth difference of three reference objects with known coordinates (Fig. B. 1(b)). The sample site's coordinates (*x*, *y*) on the street map could be obtained by solving

$$\begin{cases} \dfrac{(x_1 - x)(x_2 - x) + (y_1 - y)(y_2 - y)}{\sqrt{(x_1 - x)^2 + (y_1 - y)^2}\sqrt{(x_2 - x)^2 + (y_2 - y)^2}} - \cos\varphi_1 = 0 \\ \dfrac{(x_3 - x)(x_2 - x) + (y_3 - y)(y_2 - y)}{\sqrt{(x_3 - x)^2 + (y_3 - y)^2}\sqrt{(x_2 - x)^2 + (y_2 - y)^2}} - \cos\varphi_2 = 0 \end{cases}, \tag{b3}$$

where (x_1, y_1), (x_2, y_2), and (x_3, y_3) are the coordinates of reference points. The orientation error could be obtained by

$$\varphi_e = \varphi_1 - \arctan\left(\frac{x_1 - x}{y_1 - y}\right). \tag{b4}$$

For a panoramic camera with an orientation fix function, it is only necessary to calibrate the orientation errors at the beginning of panorama videos. In this study, all panoramas were rotted to take the south as the middle of the image.

Errors can also be caused by the road slope but can be automatically calibrated by the camera's horizon leveling function. It is also possible to calibrate the error of zenith direction in view of the parallel perspective in cube projection. The error array (φ_e, θ_e) of the zenith direction in the camera coordinates could be obtained by

$$\varphi_e = \text{atan}\left(w_f / 0.5R\right) \tag{b5}$$

and

$$\theta_e = \text{atan}\left(\frac{h_f}{\sqrt{(0.5R)^2 + {w_f}^2}}\right), \tag{b6}$$

where (w_f, h_f) is the image position of the perspective focus on the face of the cube projection, and R is the edge length of the cube (Fig. B. 2). The zenith direction could be calibrated through projection transformation as described in the following section.

Appendix C. Potential reflection region classification

According to the facade pattern, the reflection types of sunlight are divided into full reflection, fragmental reflection and masked reflection (Shih & Huang, 2001). In view of the number and image area of reflective objects in the street view, only high-rise facades that occupy more 50% of the area of GCWs are considered. For facades with continuous reflective surfaces (Fig. B. 3(a)), the whole facade is extracted as one segment (full reflection). For facades with non-continuous reflective surfaces (Fig. B. 3(b)), the segmentation is operated in view of the visual area on the panorama. The intervals between windows smaller than 1° (solar visual radiant) are ignored (fragmental reflection). Otherwise, sunlight from the reflective surface will be completely shaded when its trajectory moves into the unreflective part. In this situation, the unreflective part of facade should be excluded. Masked facades (Fig. B. 3(c)) are ignored in this study. These rules are also applied to the scenes when the facade is partly occluded by the objects (e.g., trees, streetlamp, and traffic signs) in street environment.

In an actual street environment, it is necessary to consider the influence of an object's occlusion to reflection. It can be divided into two types: (1) occlusion before reflection and (2) occlusion after reflection (Fig. B. 4(a)). Occlusion before reflection is caused by the occluding objects (e.g., surrounding high-rise buildings) distributed on the route of the incident sunlight (O_1 in Fig. B. 4(a)). The occluding object images appear on the surface instead of the occluded sunlight (Fig. B. 4(b)). As this study does not consider the irradiation through more than one time of reflection, it is impossible for reflection to appear on these areas. In addition, the range of the imaging area is decided by the spatial relation of viewpoint P, occluder $O1$, and surface A, so the imaging area of O_1 does not change with the variation of solar direction B. Occlusion after reflection distributes between the reflection point and viewpoint and interrupts the reflection sunlight reaching the viewpoint. This makes the reflective surface partly or completely invisible in the street view (Fig. B. 4(b)). The exclusion of the occlusion region can be operated by extracting the sky's imaging area on the visible reflective surface (Fig. B. 4(c)). These regions are called potential reflection regions in this study. Moreover, the potential reflection region is location-specific and must be extracted site by site. By applying the above approach to all visible potential reflection regions in the street view, we can obtain the reflected sunlight's spatial distribution on the panorama.

Appendix D. Panoramic projection conversion approach

For a panorama of the width, W, and height, H. The radius of the Celestial Sphere should be $R_o = W/2\pi$. For pixel (w_i, h_i) on the initial image, the corresponding resource arrays in the celestial coordinate system (φ_i, θ_i) should be

$$\varphi_i = \frac{2\pi}{W}(w_i - W/2), \tag{d1}$$

$$\theta_i = \frac{\pi}{H}(-h_i + H/2). \tag{d2}$$

Transforming the spherical coordinates to Cartesian, the three-dimensional coordinates (xyz) is

$$x_i = R_o \cos\theta_i \cos\varphi_i, \tag{d3}$$

$$y_i = R_o \cos\theta_i \sin\varphi_i, \tag{d4}$$

$$z_i = R_o \sin\theta_i, \tag{d5}$$

where x, y, and z stand for the axis of south, west, and zenith direction, respectively. With the rotation of around y-axis, the resulting Cartesian coordinates should be

$$\begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix}. \tag{d6}$$

Accordingly, the resulting pixel in the panorama with projection transformation can be obtained as follows:

$$w_r = \frac{W}{2\pi}\varphi_r + W \Big/ 2, \tag{d7}$$

$$h_r = -\frac{H}{\pi}\theta_r + H \Big/ 2, \tag{d8}$$

where φ_r, θ_r are

$$\theta_r = \arctan 2\left(\frac{z_r}{\sqrt{{x_r}^2 + {y_r}^2}}\right), \tag{d9}$$

$$\varphi_r = \arctan\left(\frac{y_r}{x_r}\right). \tag{d10}$$

Appendix E. Standardization the non-uniform solar motion on panorama

Polar location is a special case where the solar trajectories of different dates are equal-length lines parallel to horizontal lines (Fig. E. 1). The top and button (occluded by the earth) edges of the solar apparent area are solar trajectories on the solstice. The altitudes of these two edges equal the obliquity of the ecliptic. The one-day trajectory uniformly moves between summer solstice and winter solstice with the changes of date. As the sun moves at a constant speed along the horizontal direction of the panorama for one day, the time date of the sun appearing in a certain direction at the polar location can be obtained by its spatial position on the panorama.

For solar trajectories on other locations, the solar motion on the panorama is not uniform. The non-uniform solar motion on the panorama can be standardized by the reverse transformation process.

Appendix F. EQT calibration

As the orbit of the earth around the sun is an ellipse, the difference between the perihelion and aphelion causes the sunrise hour at vernal equinox and autumnal equinox to be different. Owing to this phenomenon, one-day solar paths will horizontally shift in the time distribution coordinate system Nourian (2020), and these errors can be calibrated by calculating

$$\mathrm{EQT} = 229.2(0.000075 + 0.001868\cos B - 0.032077\sin B - 0.014615\cos 2B - 0.04089\sin 2B), \tag{f1}$$

where

$$B = 2\pi(n-1)/365 \tag{f2}$$

stands for the number of days since January 1st. Fig. F. 1 shows that the difference between the maximum and the minimum is 30.65 min. These errors are solved by a horizontal shift of the row pixels according to its *EQT* value.

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